

CSE4203: Computer Graphics
Lecture - 4 (Part - A)
Geometric Transformation

Chapter – 6

Outline

- Transformation
- Linear Transformation
 - Scaling
 - Shearing
 - Rotation
- Composite Transformation

2D Linear Transformations (1/1)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

Linear Transformation: Operation of taking a **vector** and produces **another vector** by a **simple matrix multiplication**.

Scaling (1/6)

- The most basic transform is a scale along the coordinate axes.

$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

- s_x and s_y is called the **scaling factors** which determines how much scaling is applied along the x and y axis

Scaling (2/6)

- The most basic transform is a scale along the coordinate axes.

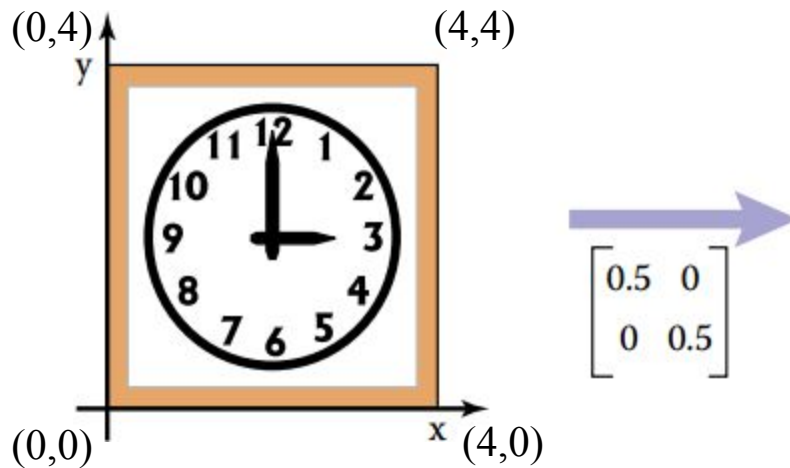
$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

The matrix does to a vector with Cartesian components (x, y):

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \end{bmatrix}$$

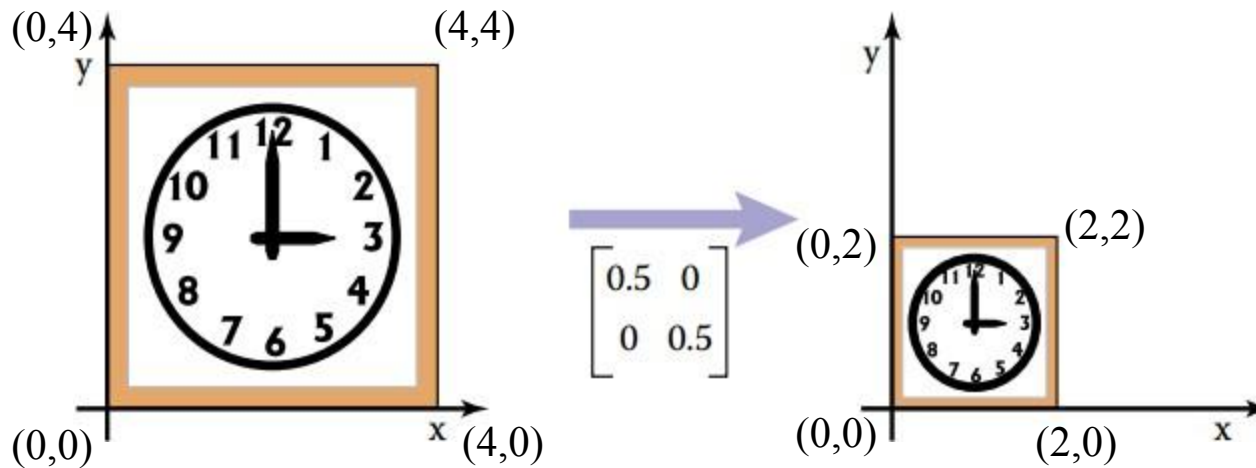
Scaling (3/6)

$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



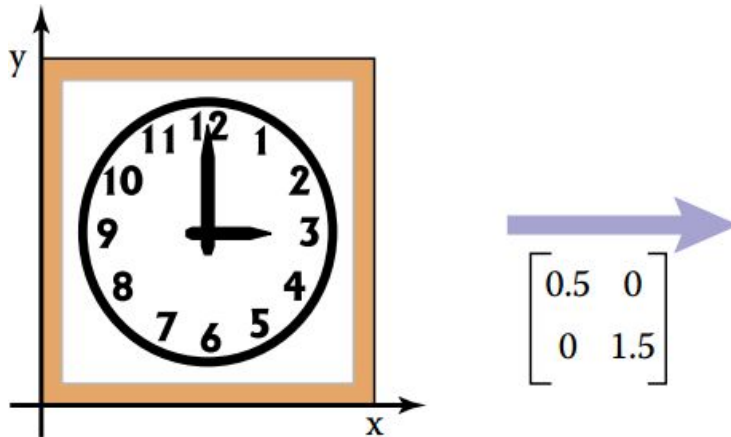
Scaling (4/6)

$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



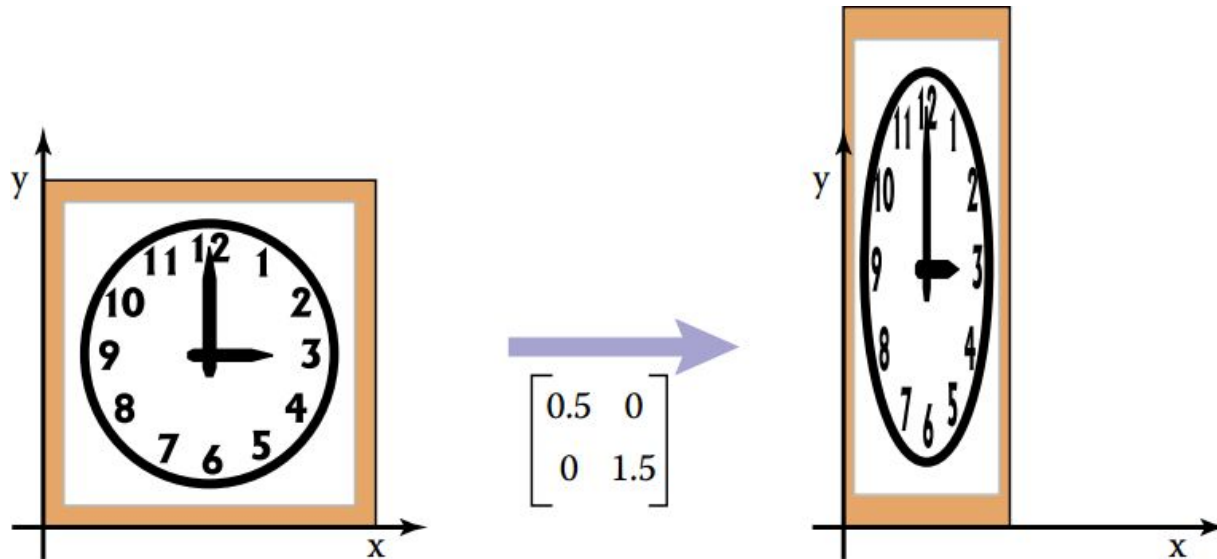
Scaling (5/6)

$$\text{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



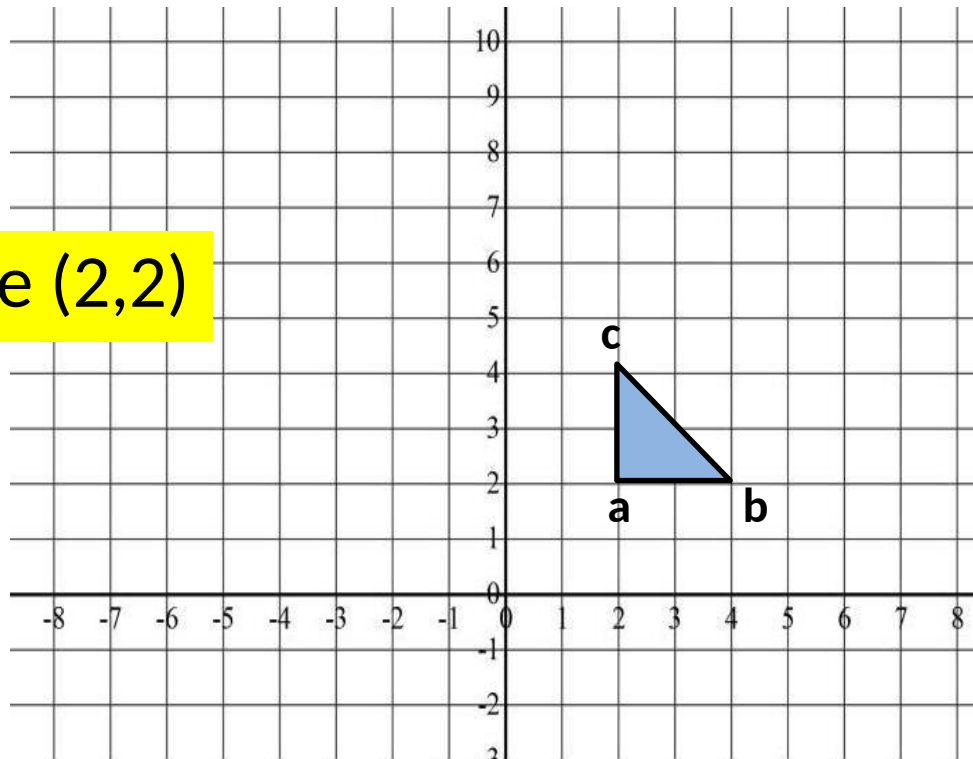
Scaling (6/6)

$$\text{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



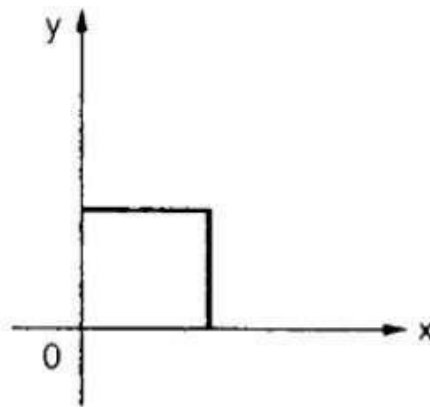
Scaling (6/6)

Scale (2,2)

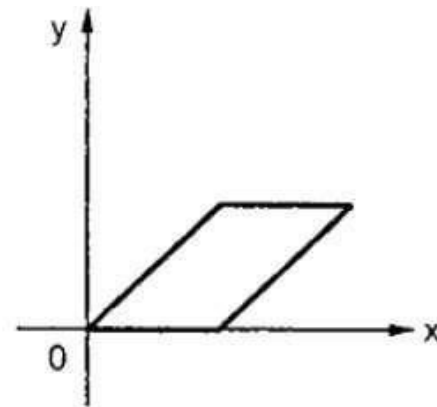


Shearing (1/5)

- A shear is something that pushes things sideways



(a) Original object



(b) Object after x shear

Shearing (2/5)

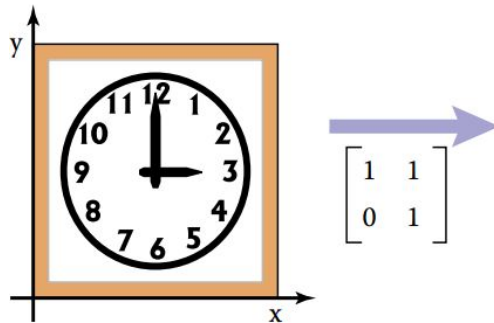
- A shear is something that pushes things sideways



$$\text{shear-x}(s) = \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}, \quad \text{shear-y}(s) = \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix}$$

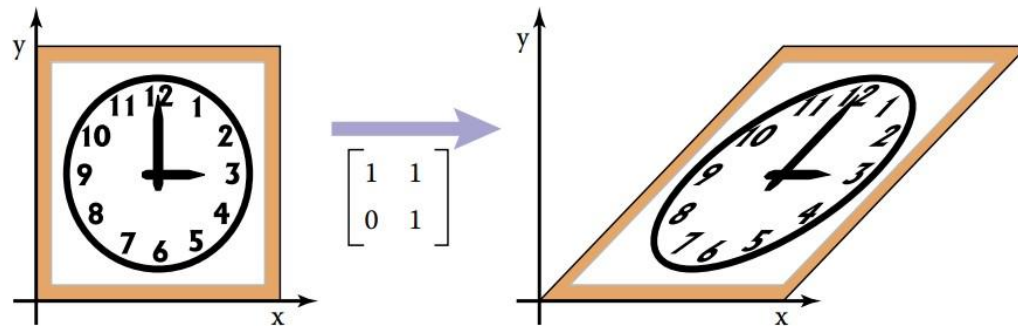
Shearing (3/5)

$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



Shearing (4/5)

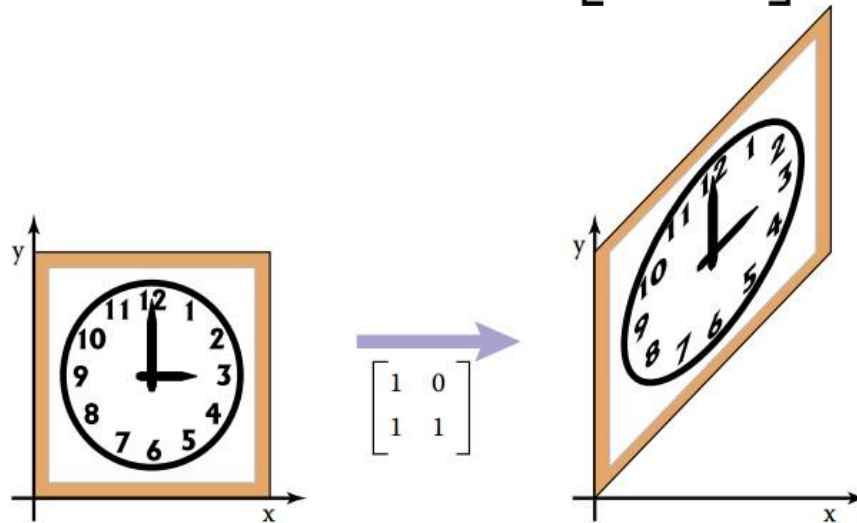
$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



An x-shear matrix moves points to the right in proportion to their y-coordinate.

Shearing (5/5)

$$\text{shear-y}(1) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

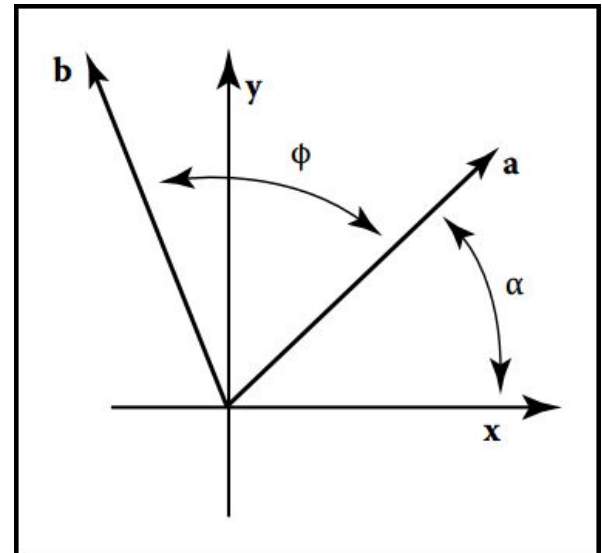


A y-shear matrix moves points up in proportion to their x-coordinate

Rotation (1/11)

$$x_a = r \cos \alpha,$$

$$y_a =$$



Rotation (2/11)

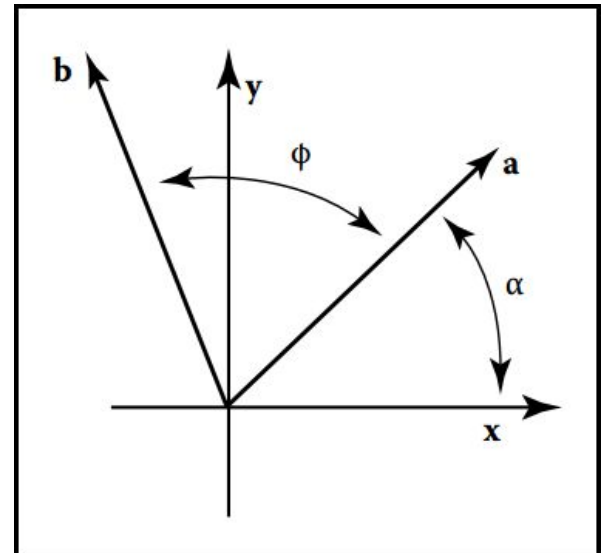
$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

$$x_b = r \cos(\alpha + \phi) =$$

$$y_b =$$

Anti-clockwise rotation = (+) rotation
Clockwise rotation = (-) rotation



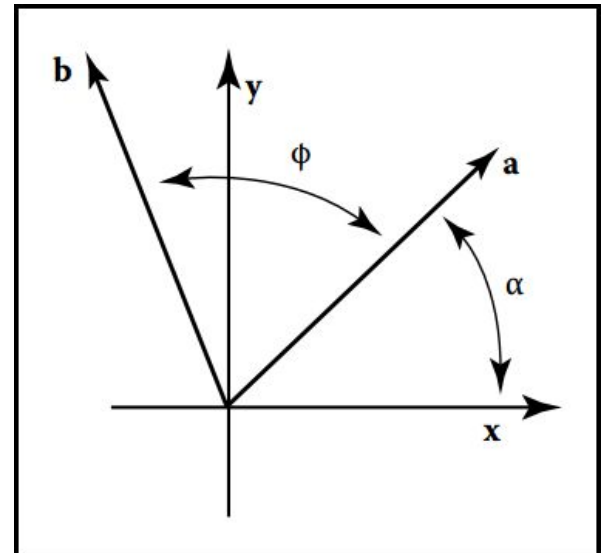
Rotation (3/11)

$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b =$$



Rotation (4/11)

$$x_a = r \cos \alpha,$$

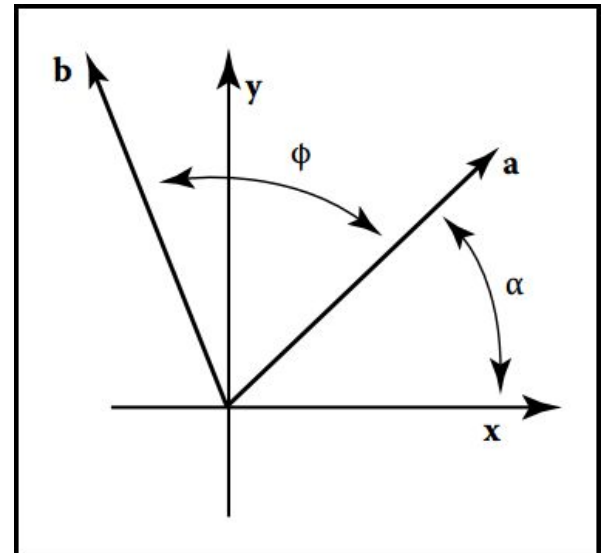
$$y_a = r \sin \alpha.$$

$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b = r \sin(\alpha + \phi) = r \sin \alpha \cos \phi + r \cos \alpha \sin \phi.$$

$$x_b =$$

$$y_b =$$



Rotation (5/11)

$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

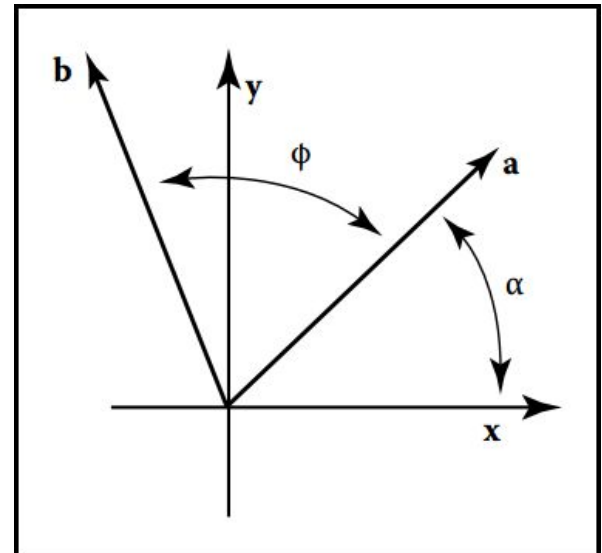
$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b = r \sin(\alpha + \phi) = r \sin \alpha \cos \phi + r \cos \alpha \sin \phi.$$

$$x_b = x_a \cos \phi - y_a \sin \phi,$$

$$y_b = y_a \cos \phi + x_a \sin \phi.$$

$$\text{rotate}(\phi) = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix}$$



Rotation (6/11)

$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

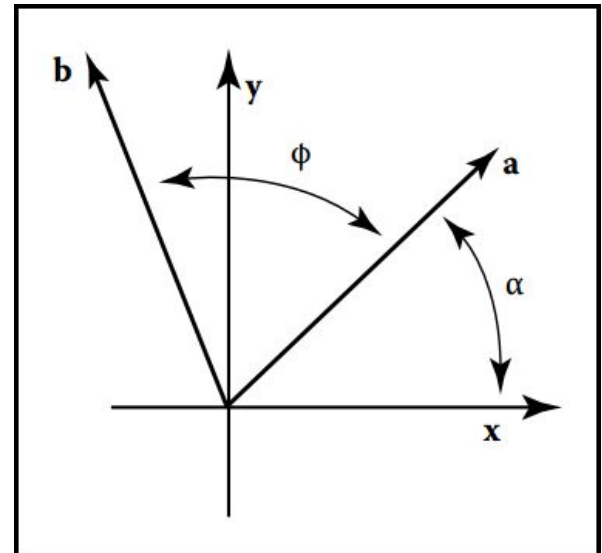
$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b = r \sin(\alpha + \phi) = r \sin \alpha \cos \phi + r \cos \alpha \sin \phi.$$

$$x_b = x_a \cos \phi - y_a \sin \phi,$$

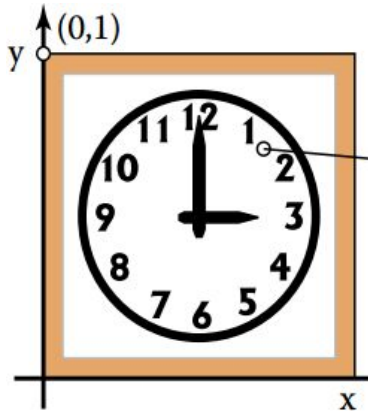
$$y_b = y_a \cos \phi + x_a \sin \phi.$$

$$\text{rotate}(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$



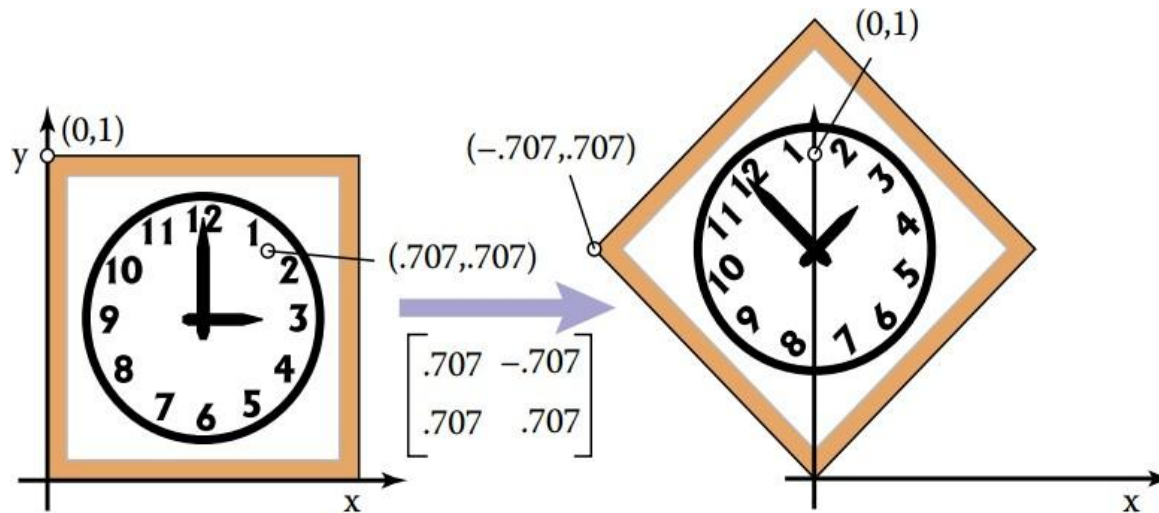
Rotation (7/11)

$$\begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$



Rotation (8/11)

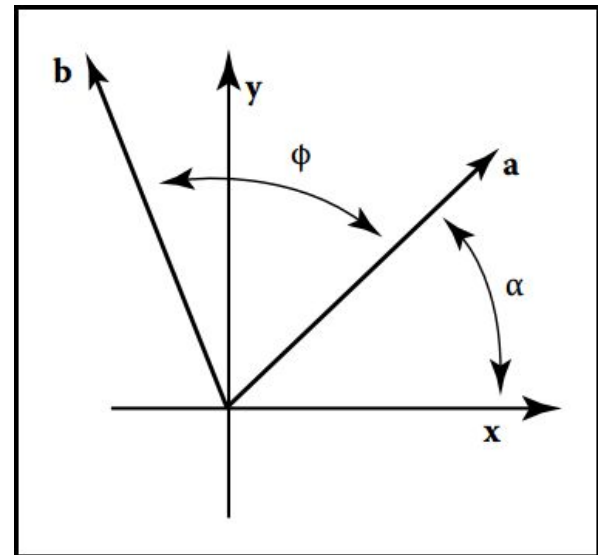
$$\begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$



Rotation (9/11)

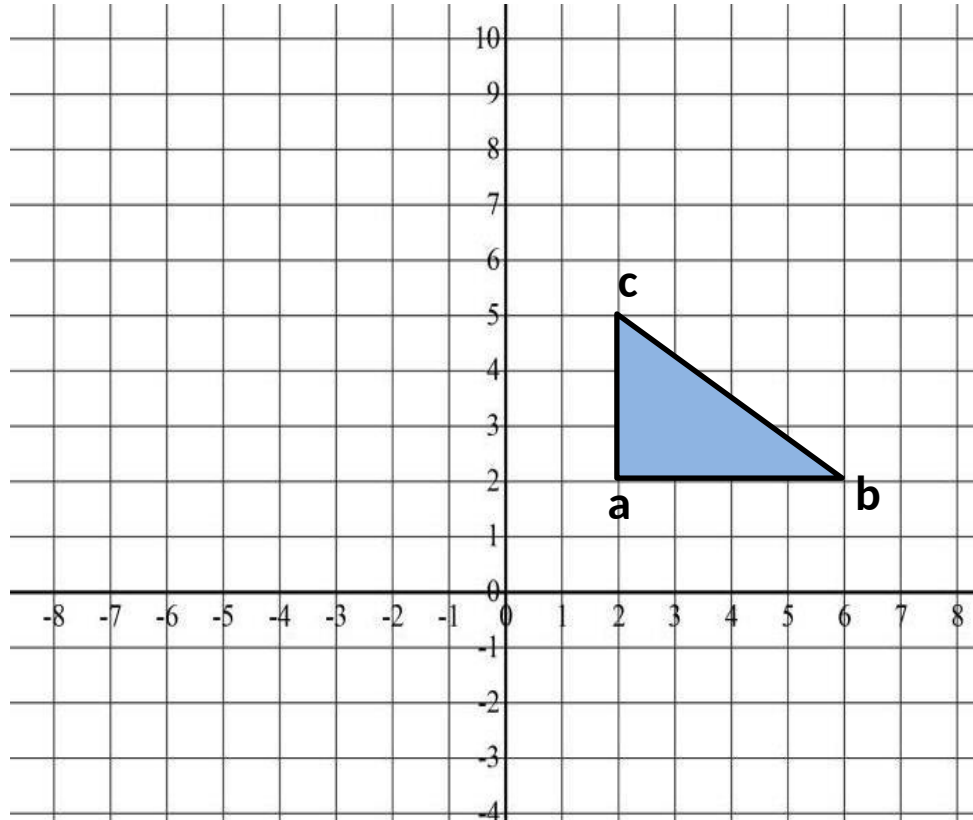
$$\text{rotate}(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

Q: What about - ve angle?



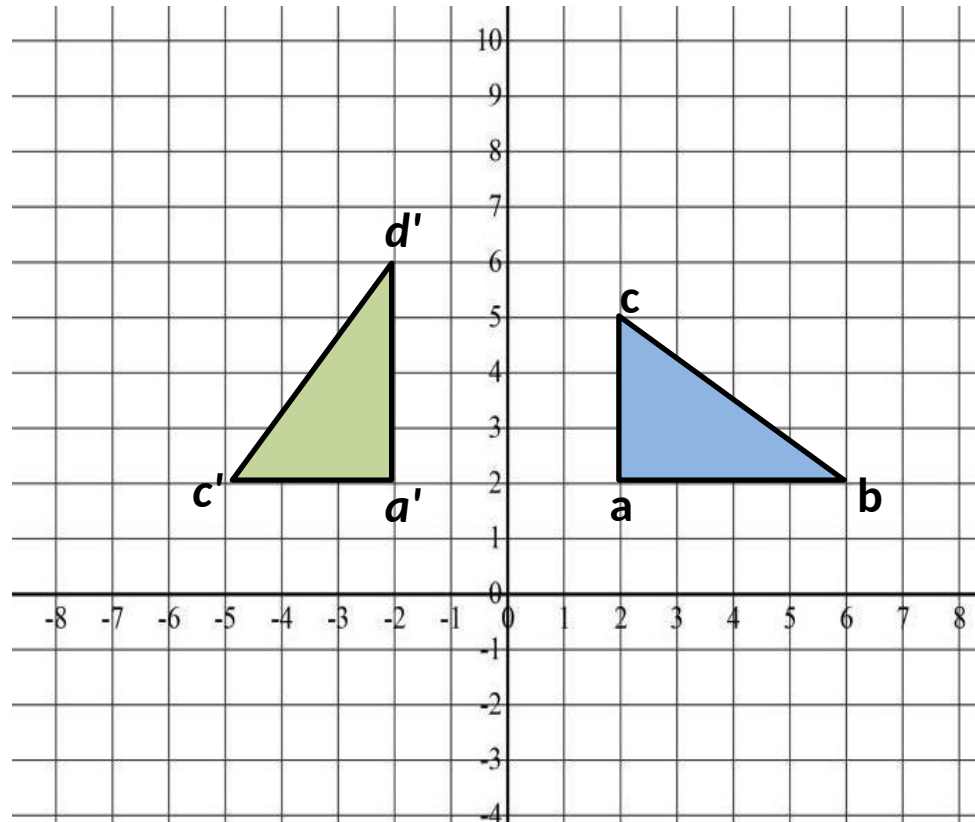
Rotation (10/11)

Rotate(90)



Rotation (11/11)

Rotate(90)

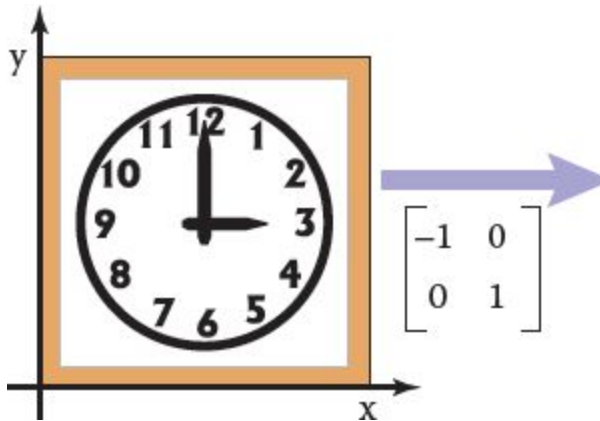


Reflection (1/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

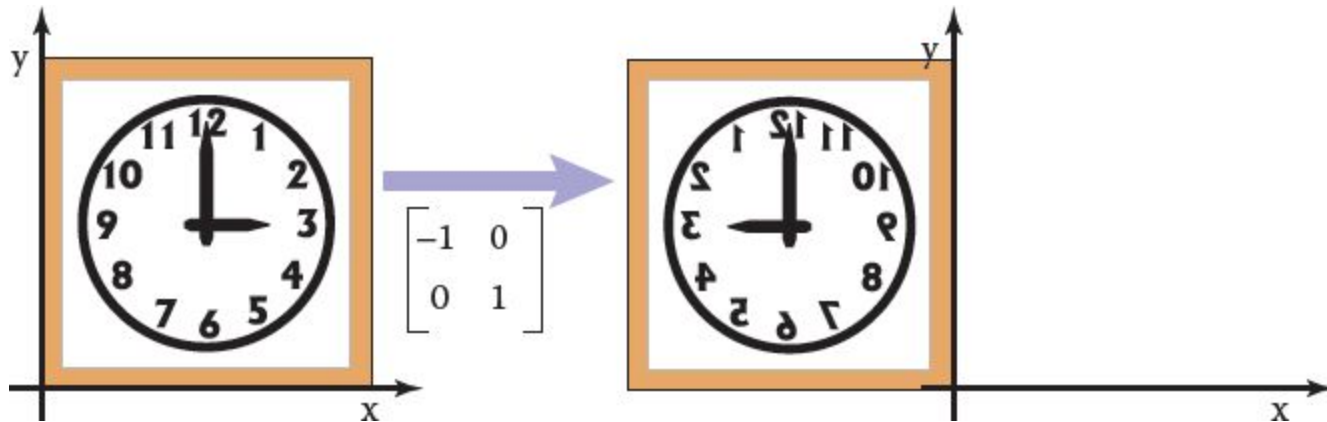
Reflection (2/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



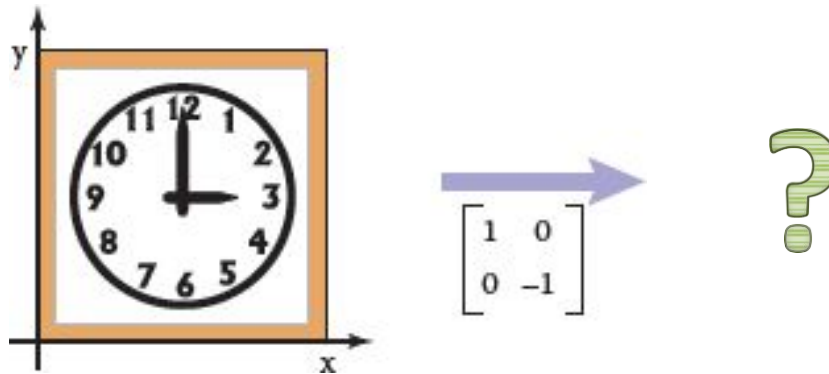
Reflection (3/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



Reflection (5/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



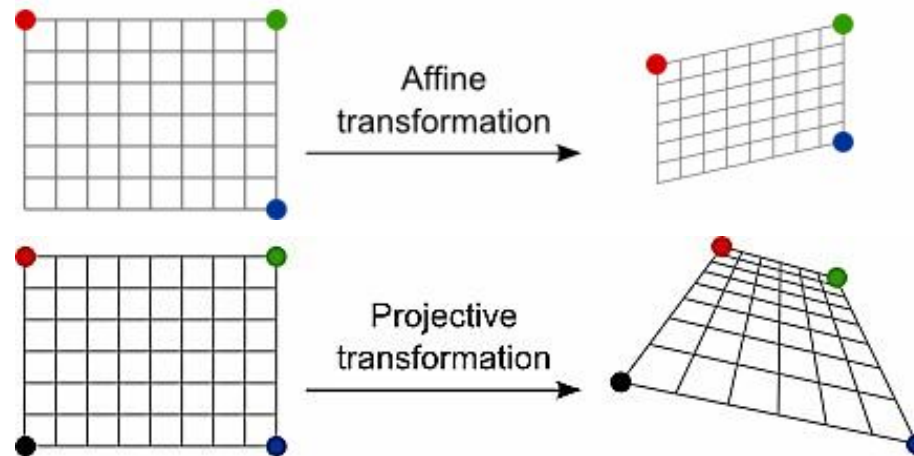
Affine transformation (1/2)

- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.

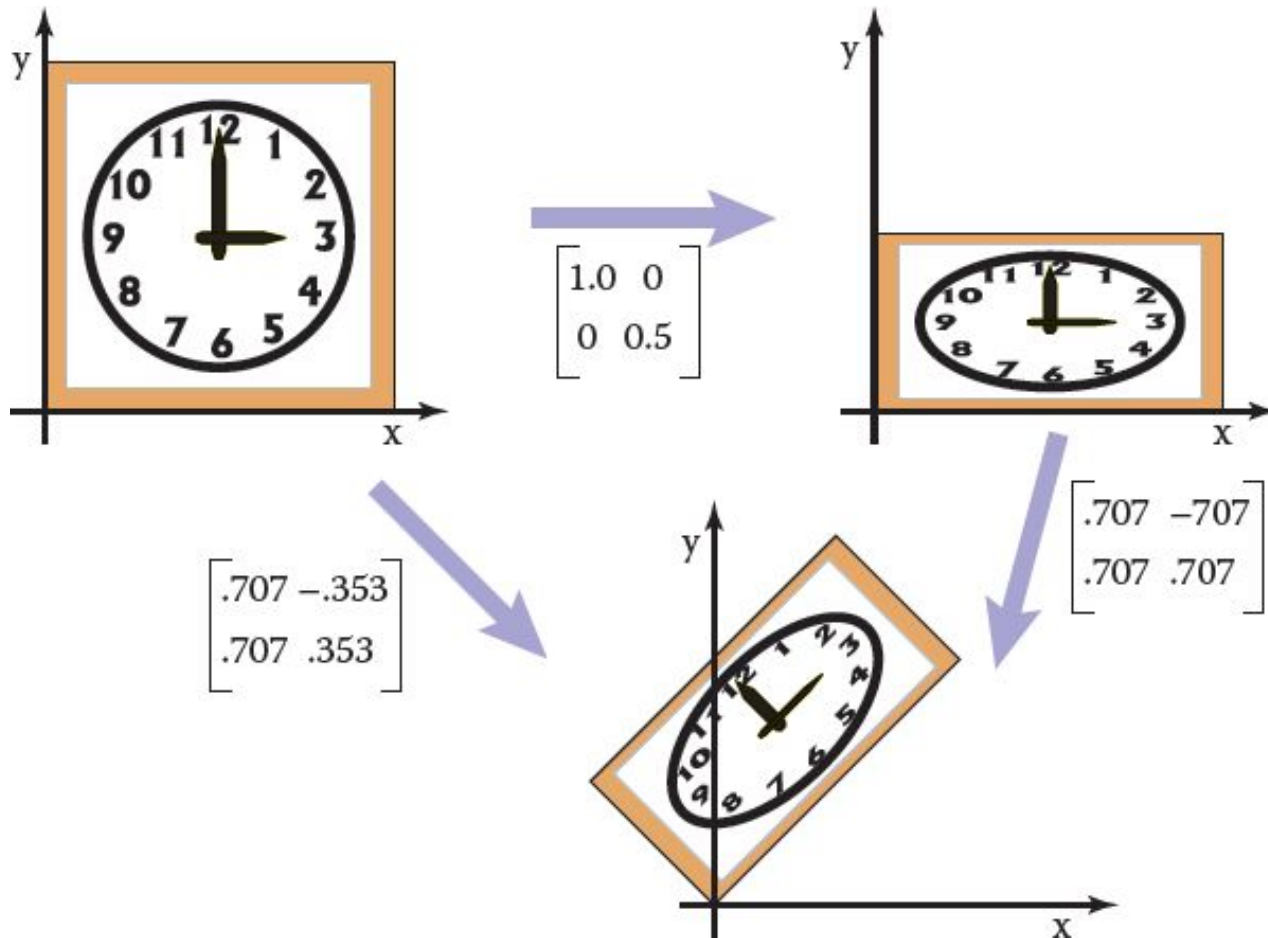


Affine transformation (2/2)

- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.



Composition of Transformations (1/11)



Composition of Transformations (2/11)

- Composite transformation is the process of combining multiple transformations
 - i.e., for a 2D point v_1 we might want to –
 1. first apply a scale S
 2. then a rotation R .
- This would be done in two steps:
 1. **first, $v_2 = S v_1$**
 2. **then, $v_3 = R v_2$.**

Composition of Transformations (3/11)

Therefore –

1. $v_2 = S v_1$

2. $v_3 = R v_2$

3. $v_3 = R (Sv_1)$

4. $v_3 = (RS) v_1$ [*matrix multiplication is associative*]

5. $v_3 = Mv_1$ [*Where $M=RS$*]

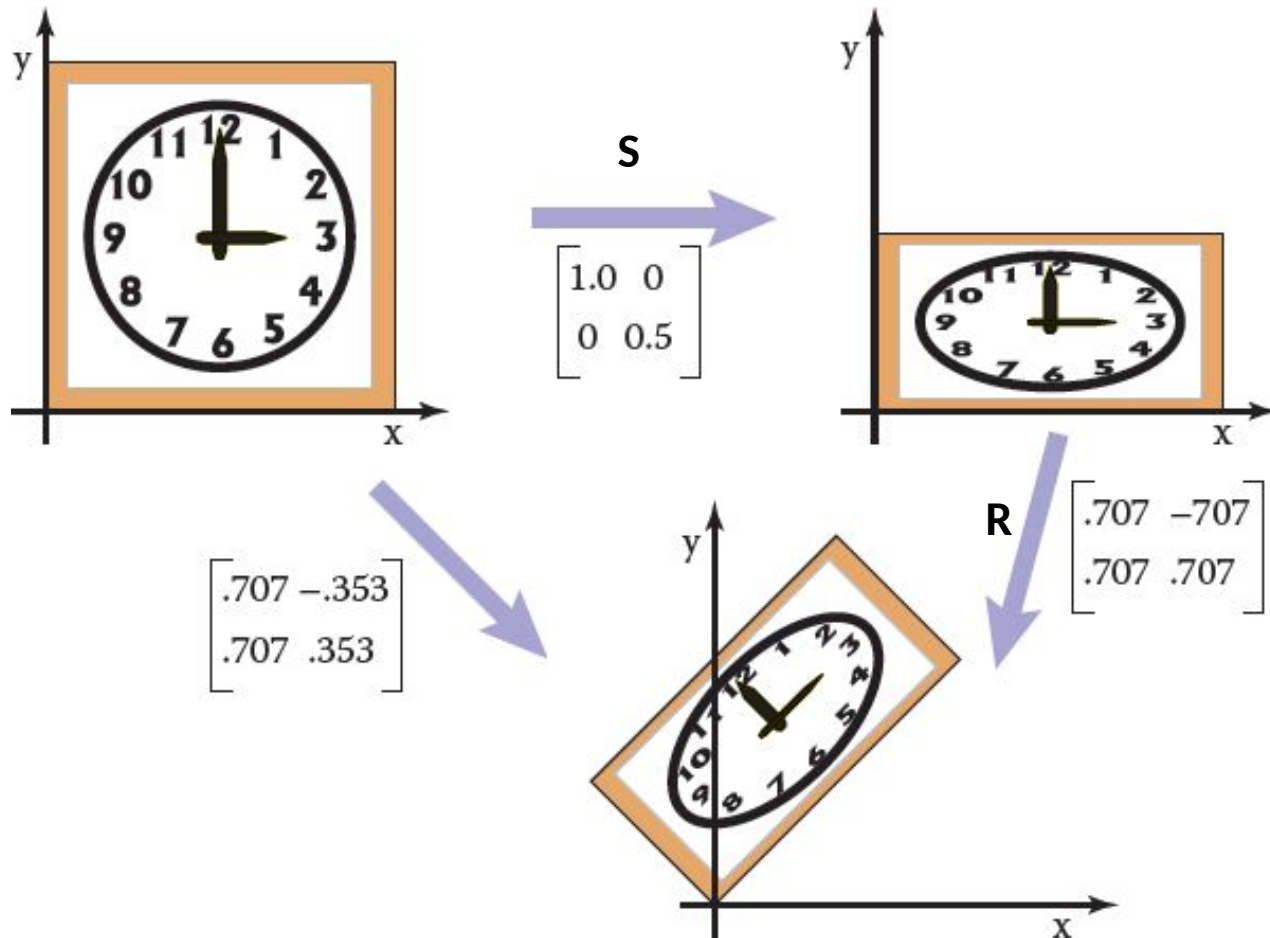
Composition of Transformations (4/11)

$$\mathbf{v}' = \mathbf{M} \mathbf{v}$$

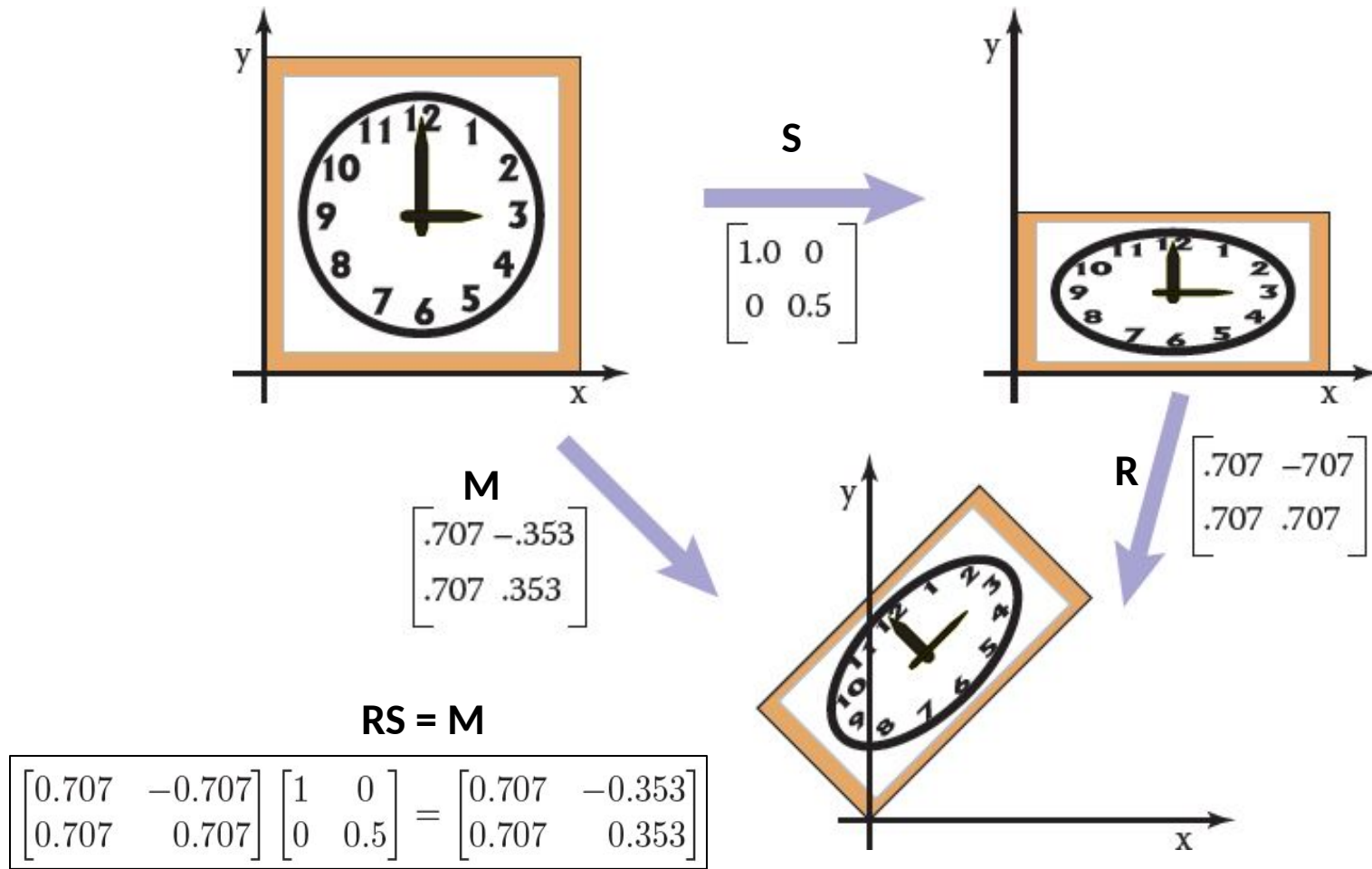
[Where $\mathbf{M} = \mathbf{R} \mathbf{S}$]

- We can represent the effects of transforming a vector by two matrices in sequence using a single matrix of the same size
 - which we can compute by multiplying the two matrices: $\mathbf{M} = \mathbf{R} \mathbf{S}$

Composition of Transformations (6/11)



Composition of Transformations (7/11)



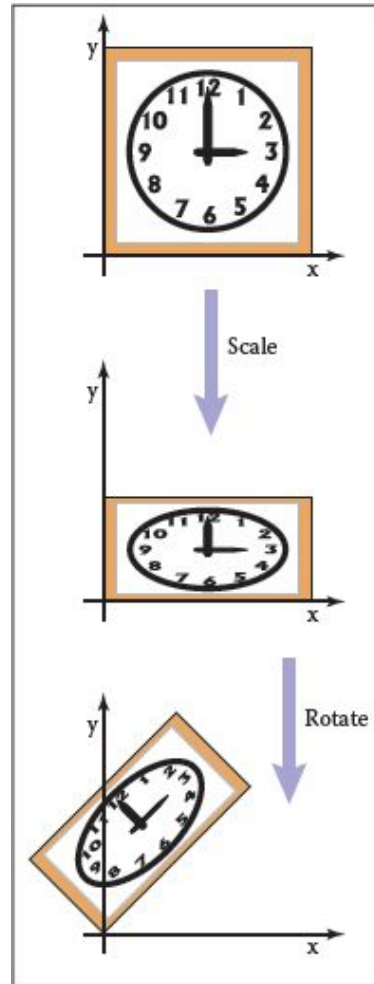
Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

Composition of Transformations (8/11)

- It is very important to remember that these transforms are applied :
 - **from the right side first.**
 - So the matrix $M = RS$
 - first applies S and then R .

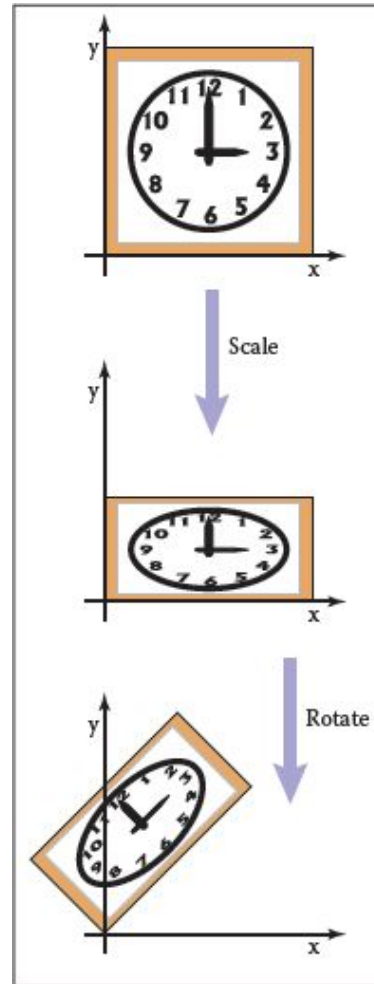
Composition of Transformations (9/11)

$$M = RS$$

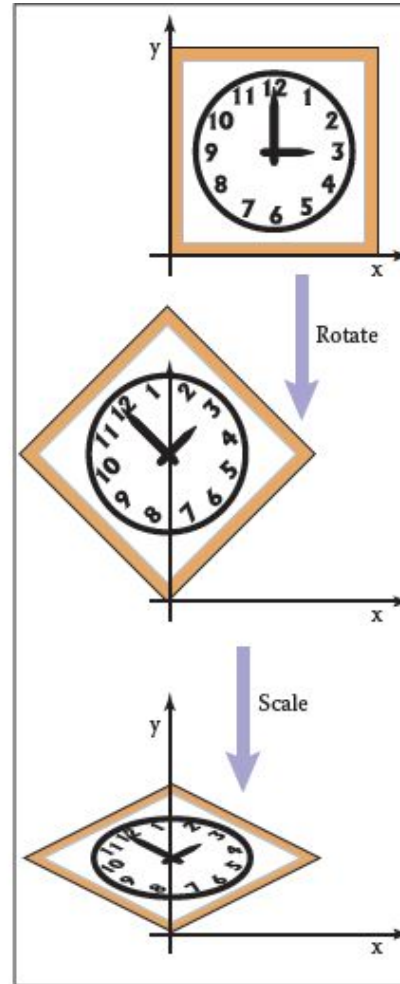


Composition of Transformations (10/11)

$$M = RS$$

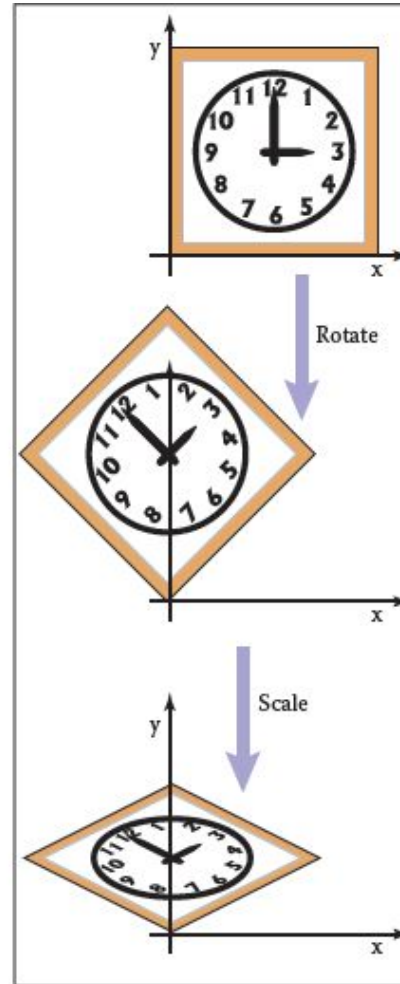
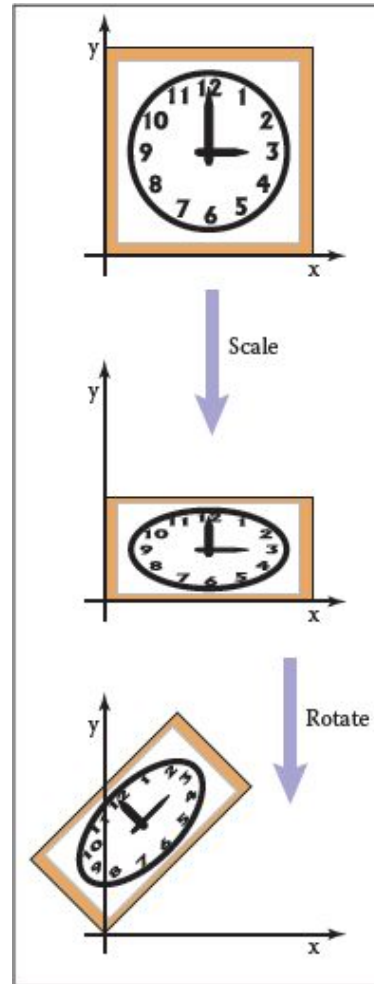


$$M = ?$$



Composition of Transformations (11/11)

$$M = RS$$

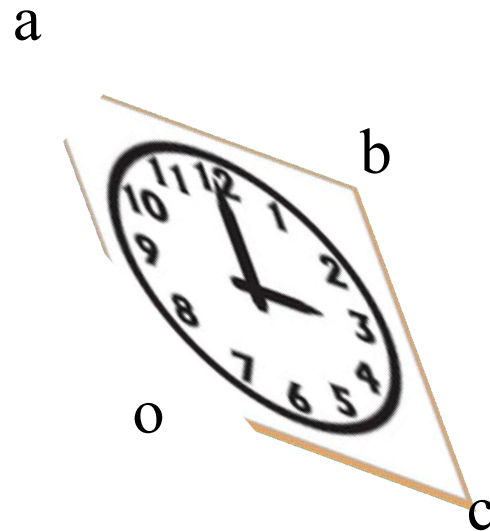
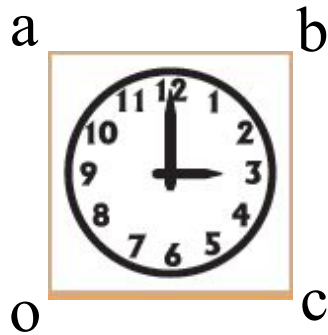


$$M =$$

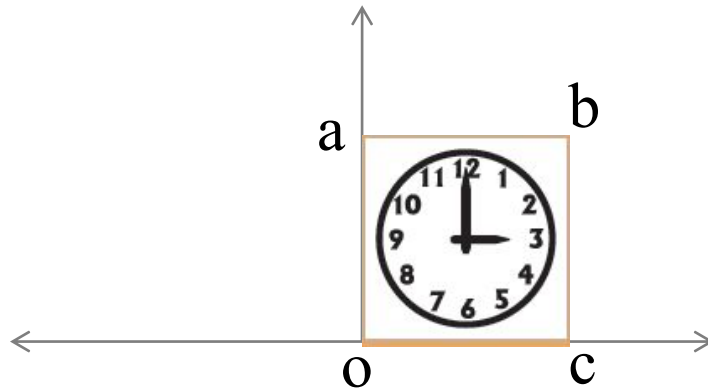
Q: What about more
than two
transformations:
 $T_1 \rightarrow T_2 \rightarrow T_3 \dots$
 $\rightarrow T_n$

Practice Problem - 1

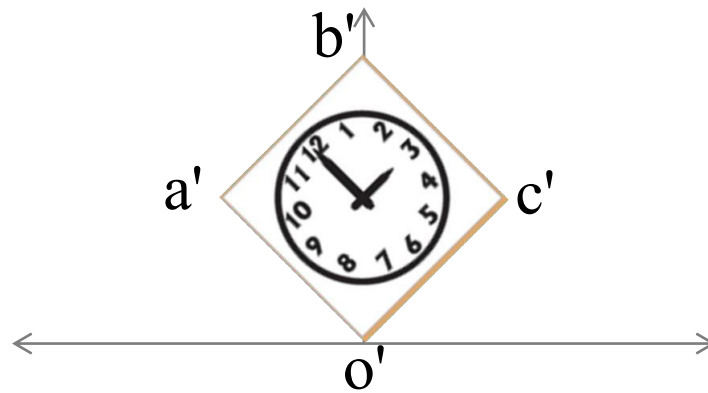
- Stretch the clock by 50% along one of its diagonals
 - so that 8:00 through 1:00 move to the northwest and 2:00 through 7:00 move to the southeast.



Practice Problem – 1 (Sol.)

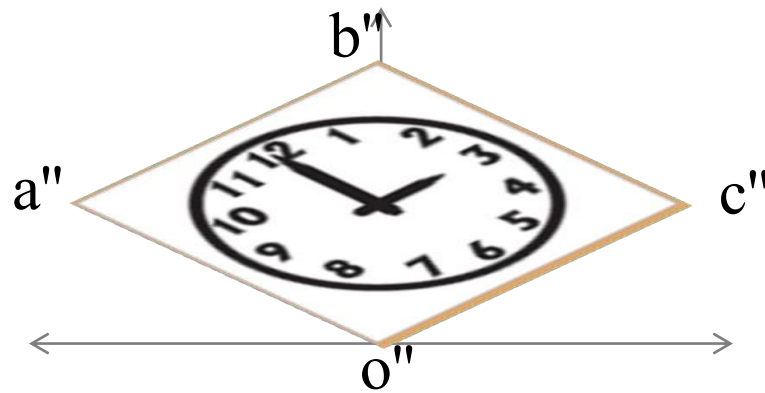


Practice Problem – 1 (Sol.)



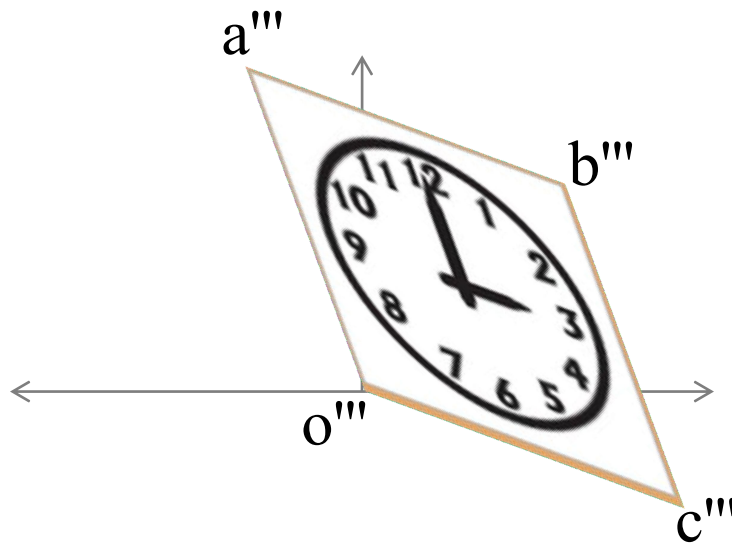
rotate(?)

Practice Problem – 1 (Sol.)



scale(?)

Practice Problem – 1 (Sol.)



rotate(?)

Practice Problem – 1 (Sol.)

- rotate(45°) $\rightarrow$ scale(1.5, 1) $\rightarrow$ rotate(-45°).

– *Q: Draw the steps*

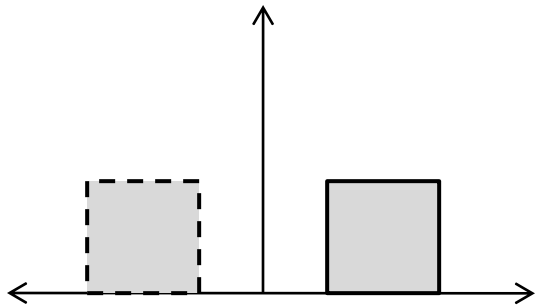
- $M = R(-45^\circ) S(1.5, 1) R(45^\circ)$
 $= R^T S R$

– *Q: Calculate the matrix*

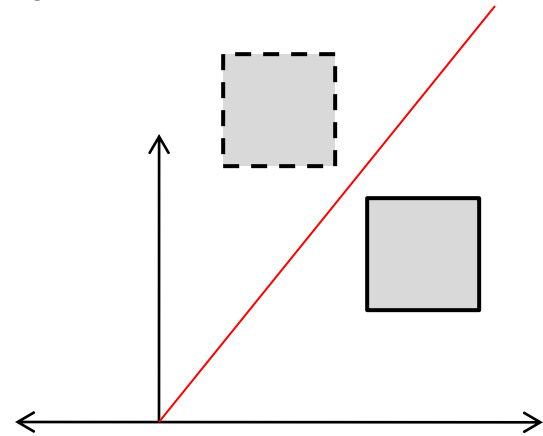
Practice Problem – 2

- Reflect the clock along a line goes through origin:

$$y = mx + c$$



w.r.t y-axis



w.r.t arbitrary line

Further Reading

- Fundamentals of Computer Graphics, 4th Edition -
Chapter 6

Thank you

CSE4203: Computer Graphics
Lecture – 4 (part - B)
Transformation Matrices

Outline

- 3D Linear Transformation
- 3D Scaling
- 3D Rotation
- Translation
- Affine Transformation

3D Transformation (1/1)

- The linear 3D transforms are an extension of the 2D transforms.

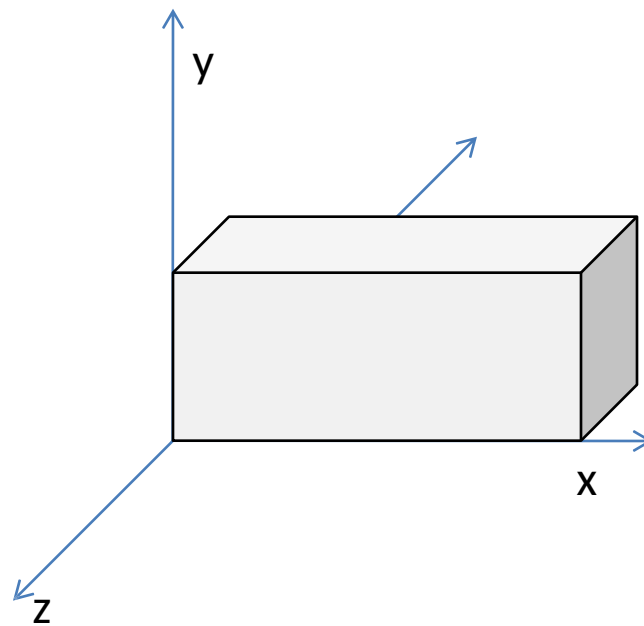
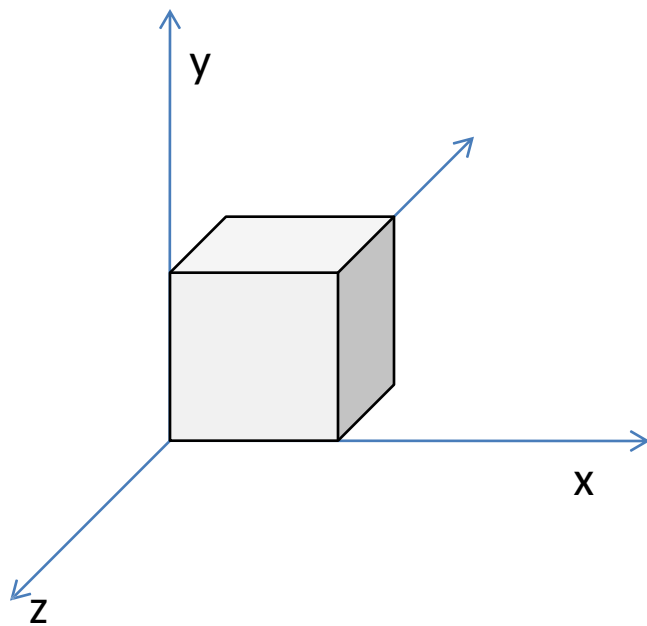
- For 2D:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

- For 3D:

3D Scaling (1/1)

$$\text{scale}(s_x, s_y, s_z) = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}$$



3D Rotation (1/5)

- Rotation around axis
 - Counter-clockwise, w.r.t rotation axis.

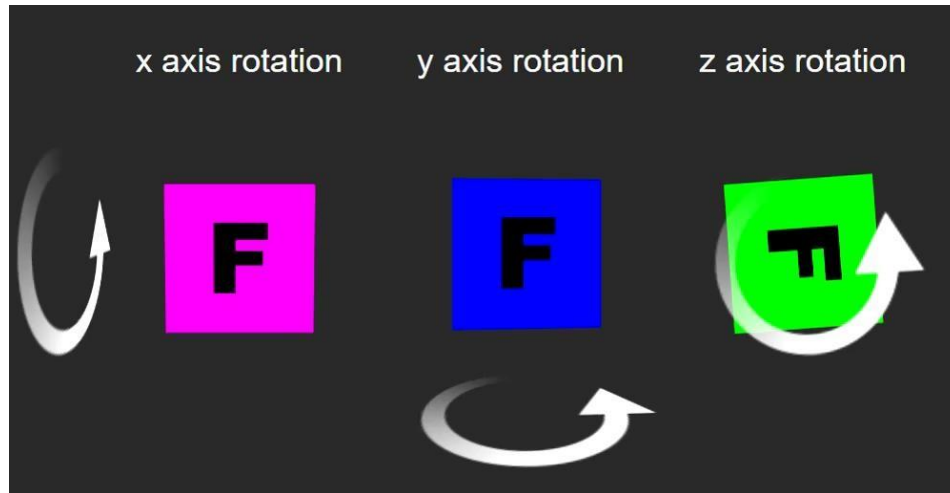
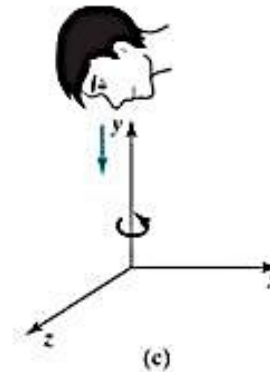
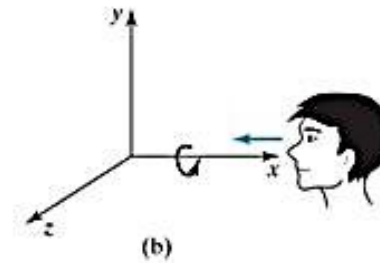
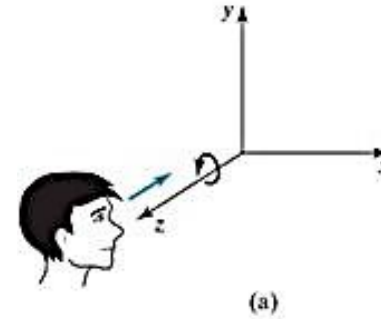
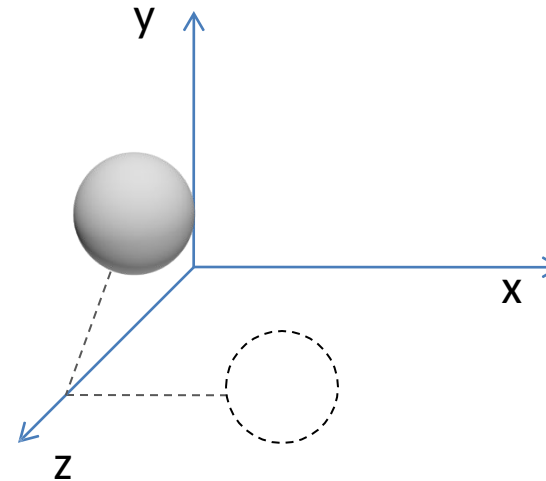
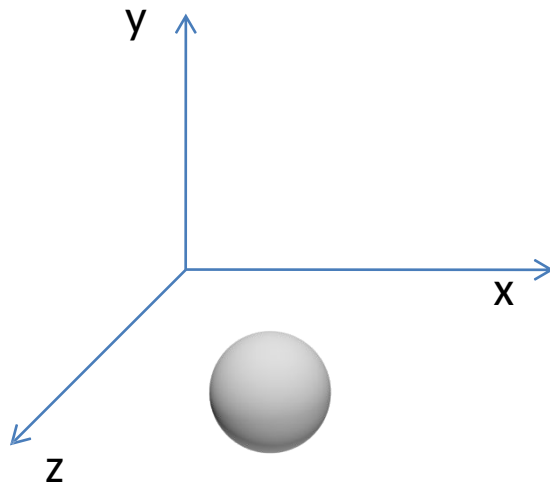


Image Source: <https://slideplayer.com/slide/4889962/>



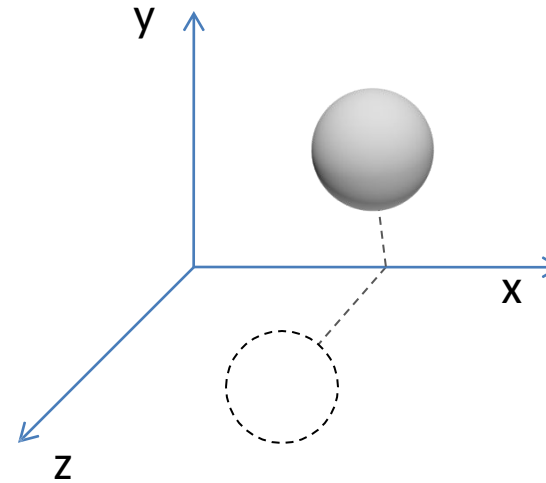
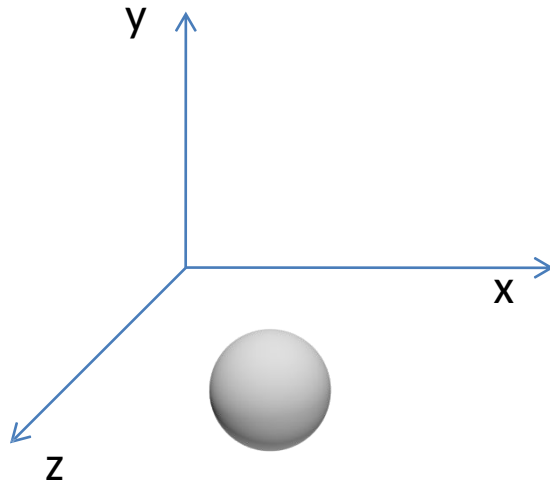
3D Rotation (2/5)

$$\text{rotate-z}(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



3D Rotation (3/5)

$$\text{rotate-x}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$



3D Rotation (4/5)

$$\text{rotate-z}(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{rotate-x}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

$$\text{rotate-y}(\phi) = \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}$$

3D Rotation (5/5)

$$\text{rotate-z}(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{rotate-x}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

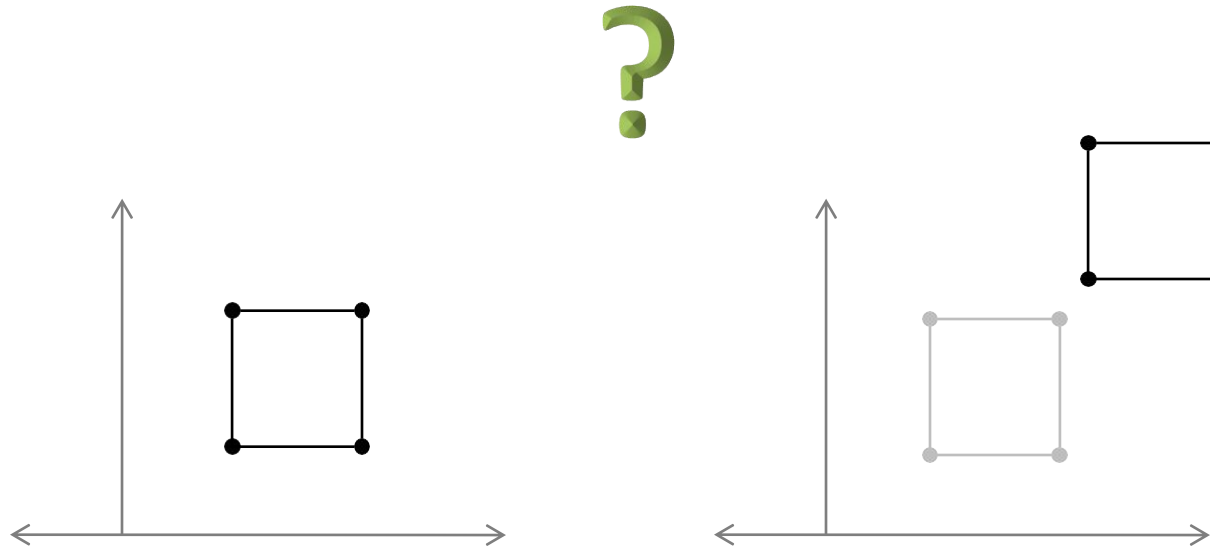
$$\text{rotate-y}(\phi) = \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}$$

Q: Why is it different?*

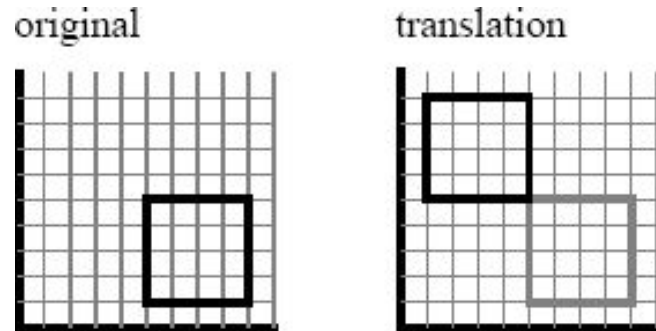
* <https://robotics.stackexchange.com/questions/10702/rotation-matrix-sign-convention-confusion>

Translation in 2D (1/8)

- Move or Translate to another position.



Translation in 2D (3/8)



$$\begin{aligned}x' &= x + t_x \\ y' &= y + t_y\end{aligned}$$

$$\mathbf{v}' = \mathbf{v} + \mathbf{t}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \end{bmatrix} + \begin{bmatrix} x \\ y \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{\mathbf{v}' = \mathbf{t} + \mathbf{v}}$

Translation in 2D (4/8)

- But, for others cases, i.e. – scaling, rotation, we changed vectors $\mathbf{v}$ using a **matrix M**.
 - In 2D, these transforms have the form: -

$\begin{aligned}x' &= m_{11}x + m_{12}y, \\y' &= m_{21}x + m_{22}y.\end{aligned}$	$\mathbf{v}' = \mathbf{M} \mathbf{v}$
---	---------------------------------------

Translation in 2D (5/8)

- We **cannot** use such transforms to **translate**, only to scale and rotate them.

$\begin{aligned} x' &= m_{11}x + m_{12}y, \\ y' &= m_{21}x + m_{22}y. \end{aligned}$	$\mathbf{v}' = \mathbf{M} \mathbf{v}$
--	---------------------------------------

Translation in 2D (6/8)

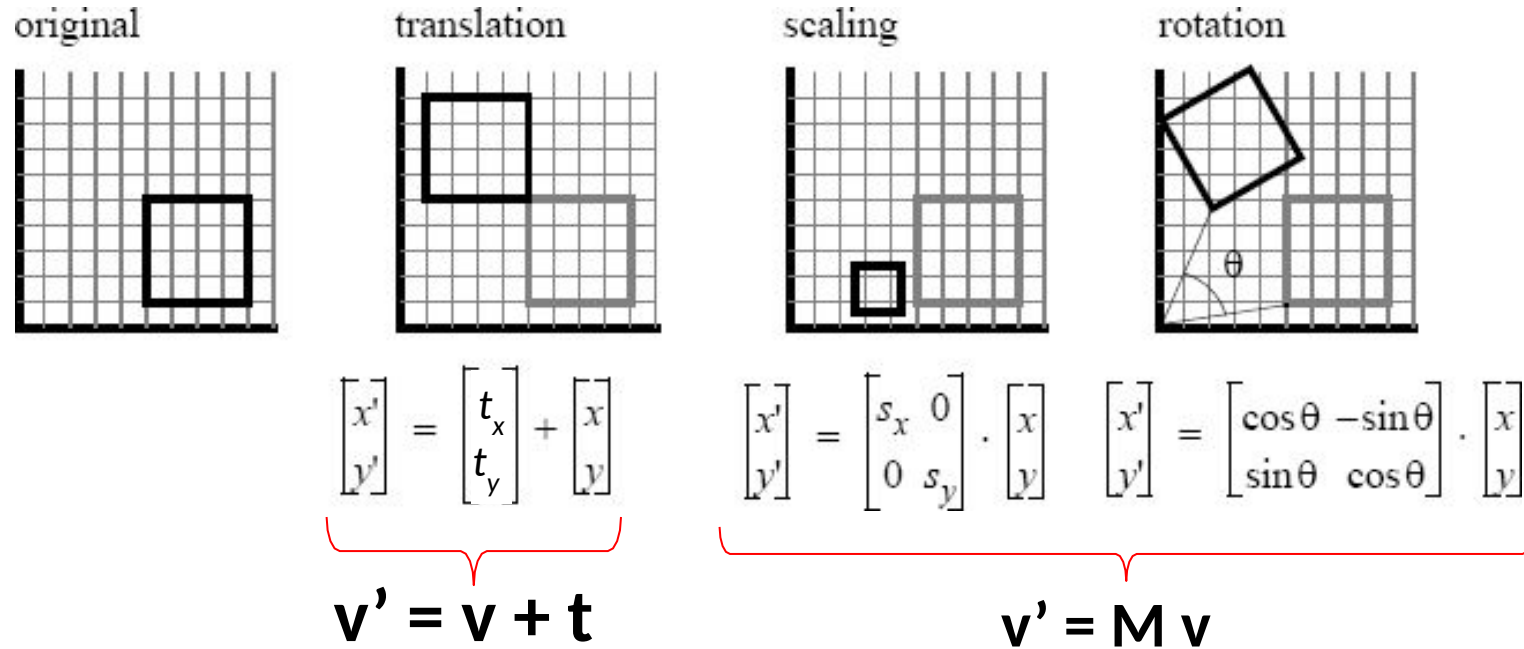
- There is just no way to do that by **multiplying** (x, y) by a 2×2 matrix.
 - **adding** translation to our system of linear transformations:

$\begin{aligned}x' &= m_{11}x + m_{12}y, \\y' &= m_{21}x + m_{22}y.\end{aligned}$	$\mathbf{v}' = \mathbf{M} \mathbf{v}$
$\begin{aligned}x' &= x + x_t, \\y' &= y + y_t.\end{aligned}$	$\mathbf{v}' = \mathbf{v} + \mathbf{t}$

Translation in 2D (7/8)

- This is perfectly feasible

—



Translation in 2D (8/8)

- This is perfectly feasible
 - But, the rule for **composing transformations** is not as simple and clean as with linear transformations.

$$T = T_n \cdot T_{n-1} \cdots T_1 \cdot T_0$$

$\begin{aligned}x' &= m_{11}x + m_{12}y, \\y' &= m_{21}x + m_{22}y.\end{aligned}$	$\mathbf{v}' = \mathbf{M} \mathbf{v}$
$\begin{aligned}x' &= x + x_t, \\y' &= y + y_t.\end{aligned}$	$\mathbf{v}' = \mathbf{v} + \mathbf{t}$

Homogeneous Coordinates (1/9)

- Instead, we can use a clever trick to get a single matrix multiplication to do both.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix}_{2 \times 2} \begin{bmatrix} x \\ y \end{bmatrix}$$

Homogeneous Coordinates (2/9)

- Instead, we can use a clever trick to get a single matrix multiplication to do both.
- The idea is simple: represent the point (x, y) by a 3D vector $[x \ y \ 1]^T$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}_{3 \times 3} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (3/9)

- Instead, we can use a clever trick to get a single matrix multiplication to do both.
- The idea is simple: represent the point (x, y) by a 3D vector $[\mathbf{x} \ y \ \mathbf{1}]^T$
- Use 3×3 matrices of the form.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & x_t \\ m_{21} & m_{22} & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (4/9)

- This kind of transformation is called an ***affine transformation***.
 - this way of implementing affine transformations by adding an extra dimension is called ***homogeneous coordinates***

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & x_t \\ m_{21} & m_{22} & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (5/9)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & x_t \\ m_{21} & m_{22} & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (6/9)

- Translation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_t \\ 0 & 1 & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (7/9)

- Scaling:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates (8/9)

- Rotation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

3D Transformation with Homogeneous Coordinates (1/1)

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

2D/ 3D Transformations (1/3)

	2D	3D
T	$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & x_t \\ 0 & 1 & y_t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$
S	$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} p & 0 & 0 & 0 \\ 0 & q & 0 & 0 \\ 0 & 0 & r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$
R	$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$	$\text{RotX} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta_x) & -\sin(\theta_x) & 0 \\ 0 & \sin(\theta_x) & \cos(\theta_x) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\text{RotY} = \begin{bmatrix} \cos(\theta_y) & 0 & \sin(\theta_y) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta_y) & 0 & \cos(\theta_y) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\text{RotZ} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) & 0 & 0 \\ \sin(\theta_z) & \cos(\theta_z) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Inverse Transformations (1/2)

T	T^{-1}
$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & 1 & r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -p \\ 0 & 1 & 0 & -q \\ 0 & 0 & 1 & -r \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$
$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} p & 0 & 0 & 0 \\ 0 & q & 0 & 0 \\ 0 & 0 & r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$	$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1/p & 0 & 0 & 0 \\ 0 & 1/q & 0 & 0 \\ 0 & 0 & 1/r & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$
$RotX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta x) & -\sin(\theta x) & 0 \\ 0 & \sin(\theta x) & \cos(\theta x) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $RotY = \begin{bmatrix} \cos(\theta y) & 0 & \sin(\theta y) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta y) & 0 & \cos(\theta y) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $RotZ = \begin{bmatrix} \cos(\theta z) & -\sin(\theta z) & 0 & 0 \\ \sin(\theta z) & \cos(\theta z) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	?

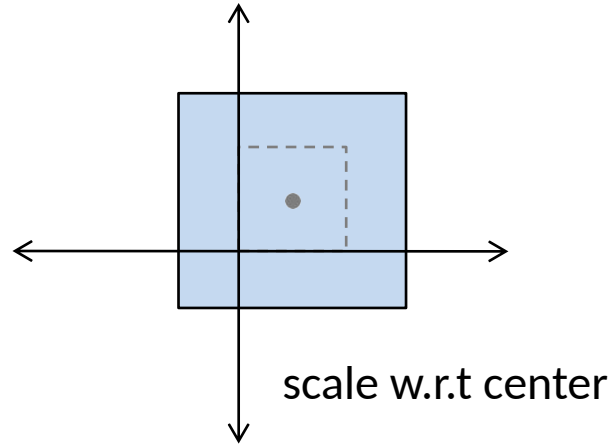
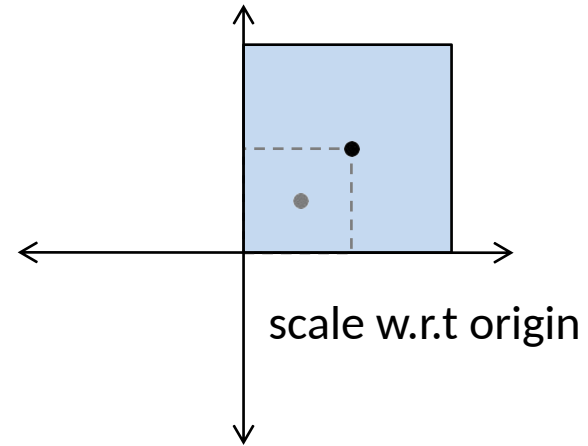
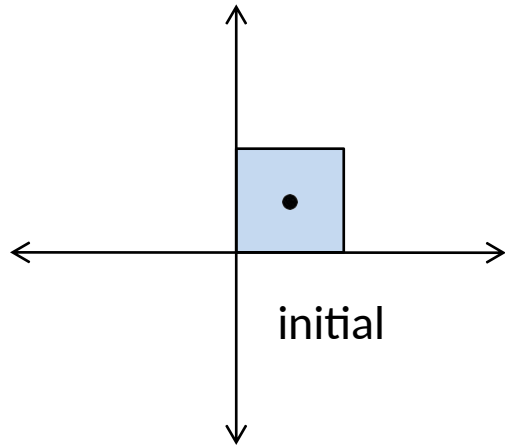
Inverse Transformations (2/2)

	Transformation	Inverse Transformation
T	$T(tx, ty, tz)$	$T^{-1} = T(-tx, -ty, -tz)$
S	$S(sx, sy, sz)$	$S^{-1} = S(1/sx, 1/sy, 1/sz)$
R	$R_x(d)$ $R_y(d)$ $R_z(d)$	$R^{-1} = R(-d) = R^T$ $R_x^{-1} = R_x^T$ $R_y^{-1} = R_y^T$ $R_z^{-1} = R_z^T$

Task: take any transformation matrix (i.e. scaling matrix S) with numerical values, do the matrix inversion and see if it becomes S^{-1}

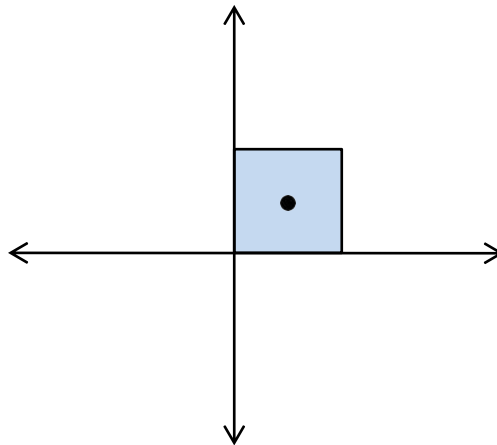
Practice Problem - 1

- Scale w.r.t the center



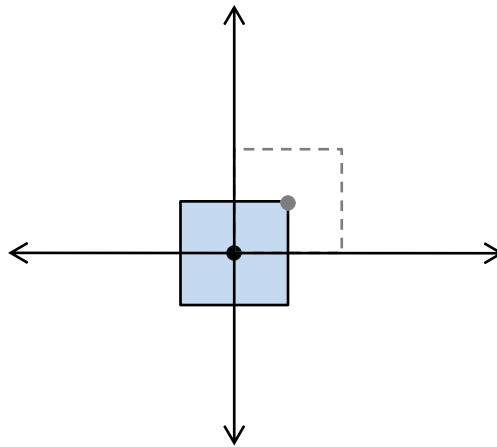
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



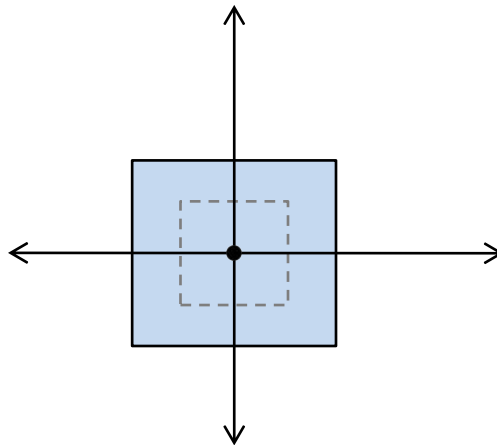
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



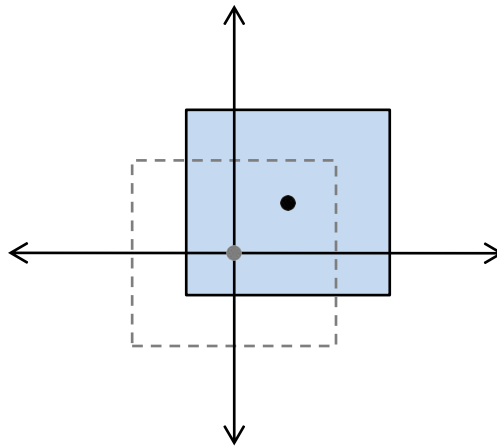
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



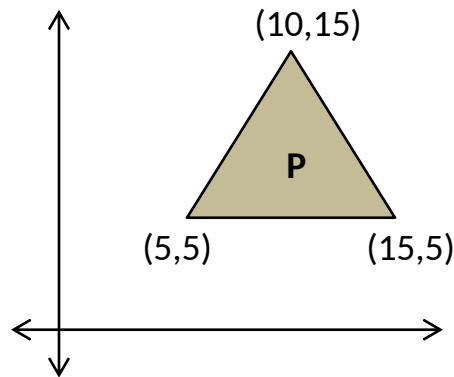
Practice Problem – 1 (Sol.)

- Scale w.r.t the center



Practice Problem - 2

- We need to rotate a pyramid **P** about point **(5, 5)** by **90°**. You have to –
 - Mention the steps to perform the task.
 - Determine the composite transformation matrix **M**.
 - Multiply **M** with **P** and determine the new coordinates **P'**.
 - Plot **P** and **P'** on the same axis to show the rotation.



Further Reading

- Fundamentals of Computer Graphics, 4th Edition - Chapter 6
(Exercise 1 – 6, 8 and 9)

CSE4203: Computer Graphics

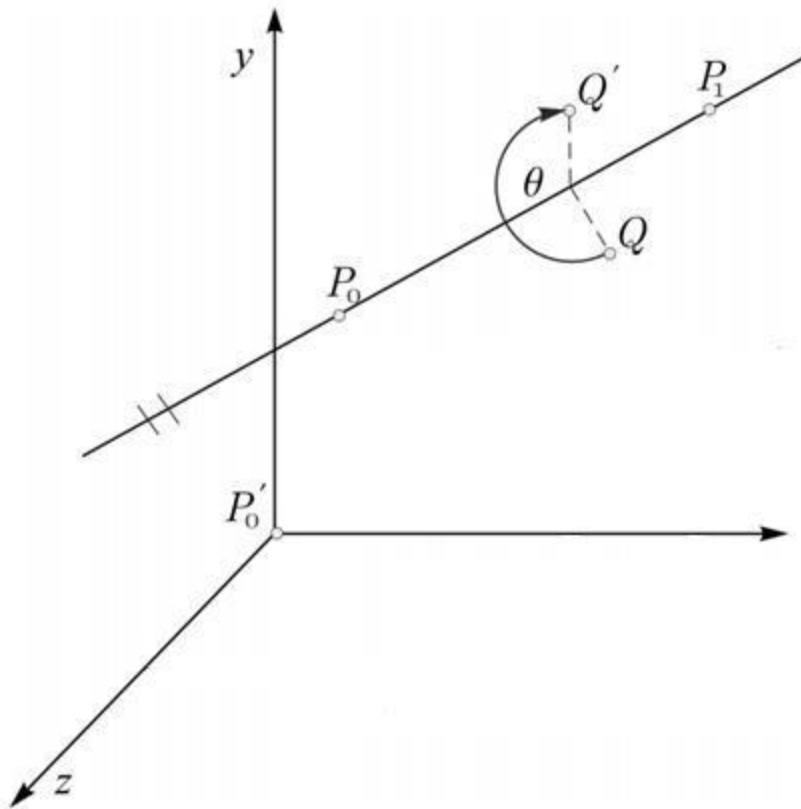
Lecture – 4 (part - C)

Transformation Matrices

Outline

- 3D Transformation
- Rotation about an arbitrary line

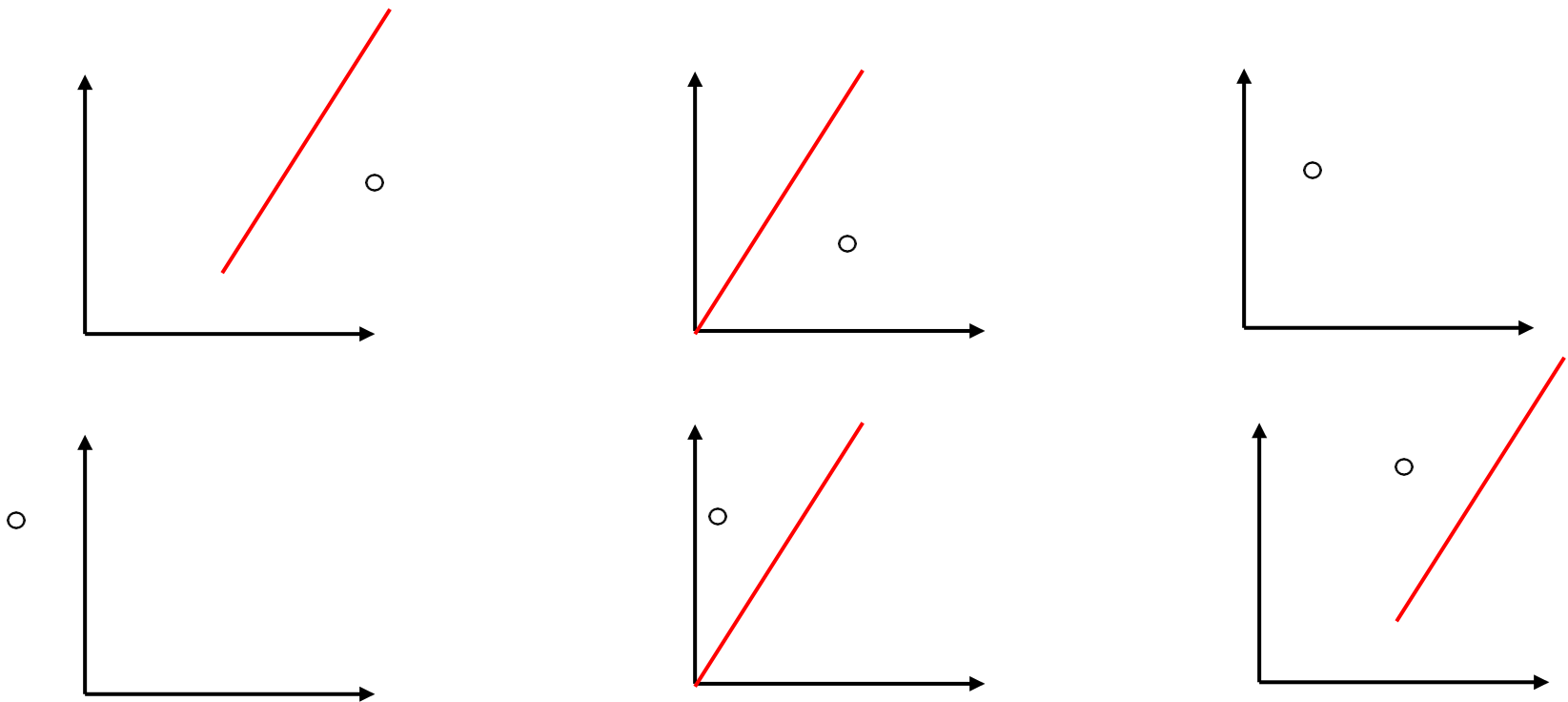
Rotation about an arbitrary line (1/1)



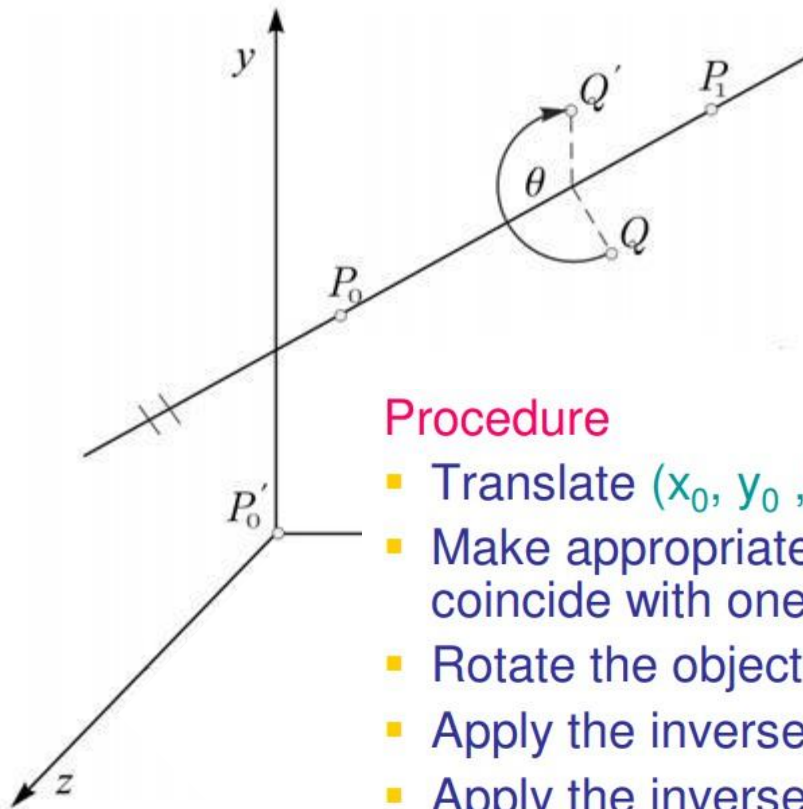
The basic idea is to make the arbitrary rotation axis coincide with one of the principle axis. Assume an arbitrary axis in space passing through the point **P0** (x_0, y_0, z_0) and **P1** (x_1, y_1, z_1) .

In 2D case (1/1)

Reflecting about an arbitrary line



Steps (1/1)

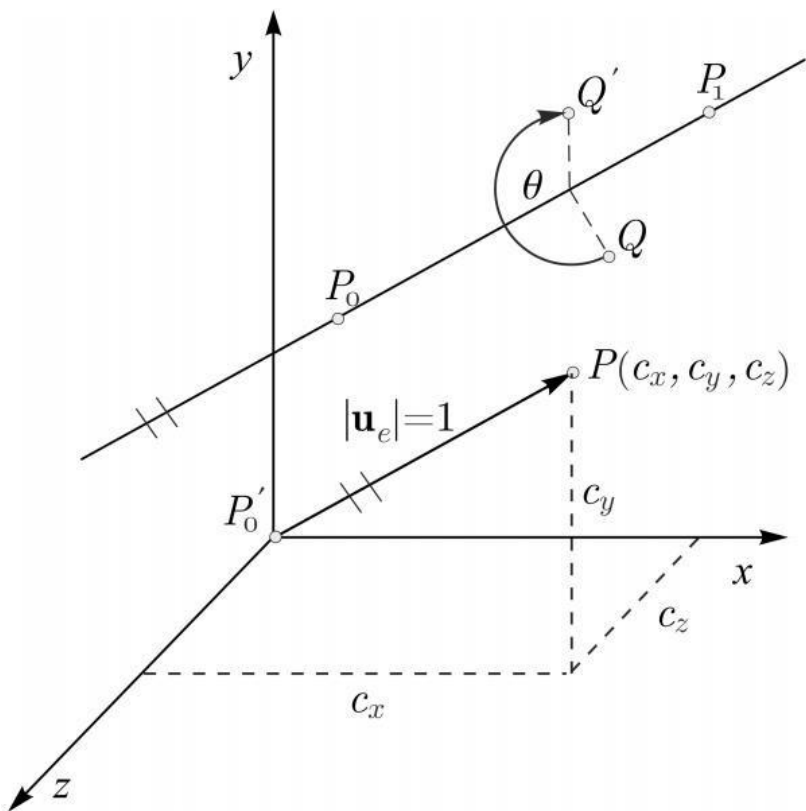


Procedure

- Translate (x_0, y_0, z_0) so that the point is at origin
- Make appropriate rotations to make the line coincide with one of the axes, say z -axis
- Rotate the object about z -axis by required angle
- Apply the inverse of step 2
- Apply the inverse of step 1

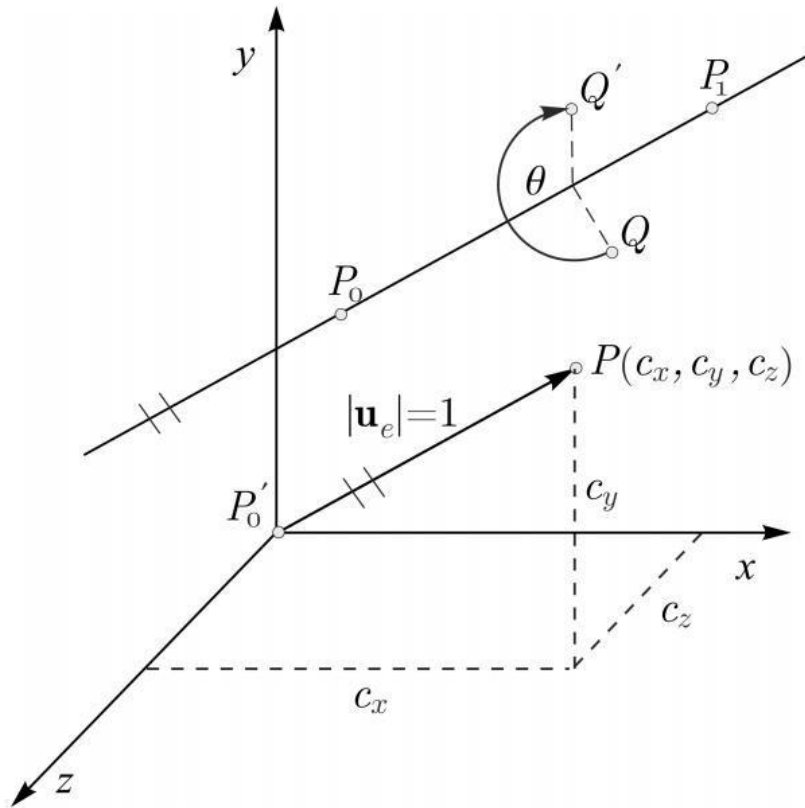
Direction Cosine (1/2)

- Direction cosines of a vector are the cosines of the angles between the vector and the three coordinate axes.



Direction Cosine (2/2)

Equivalently, they are the contributions of each component of the basis to a unit vector in that direction.



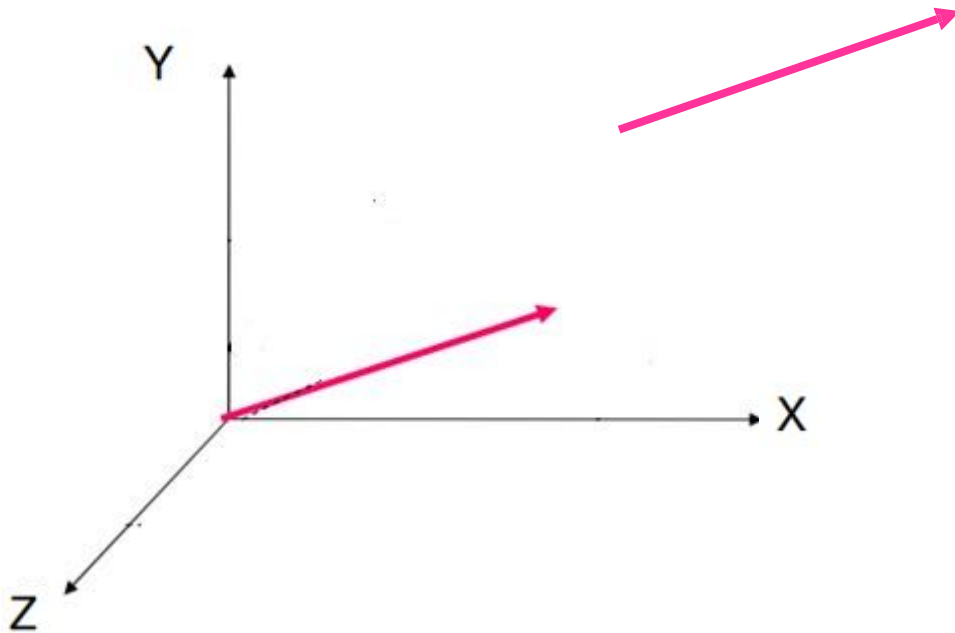
$$\mathbf{u} = P_1 - P_0$$

$$\mathbf{u}_e := \frac{\mathbf{u}}{|\mathbf{u}|} = (c_x, c_y, c_z)$$

$$c_x^2 + c_y^2 + c_z^2 = 1$$

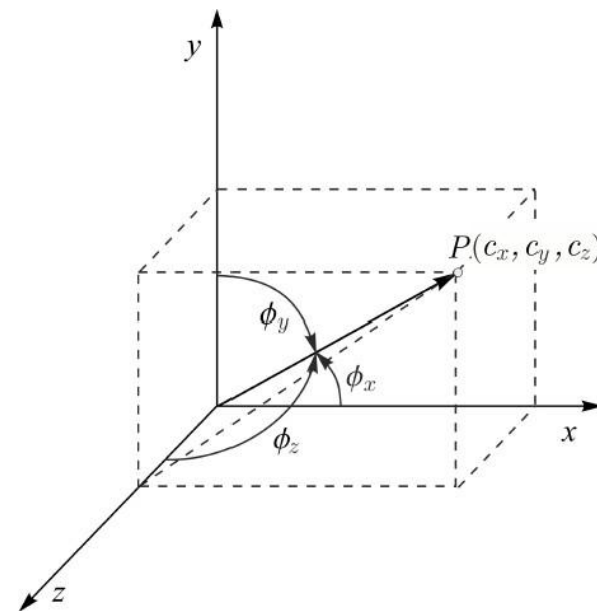
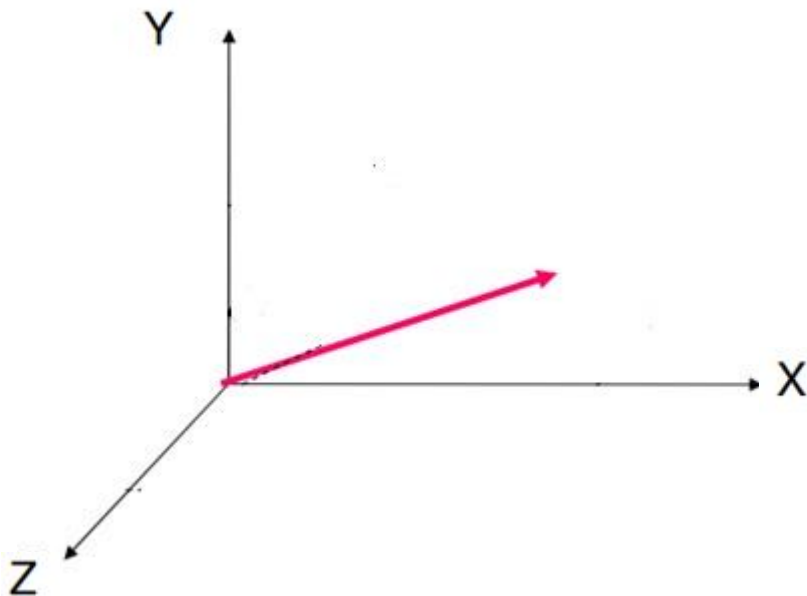
$$\cos \phi_x = c_x, \quad \cos \phi_y = c_y, \quad \cos \phi_z = c_z$$

Coinciding the line with Principal axis (1/5)



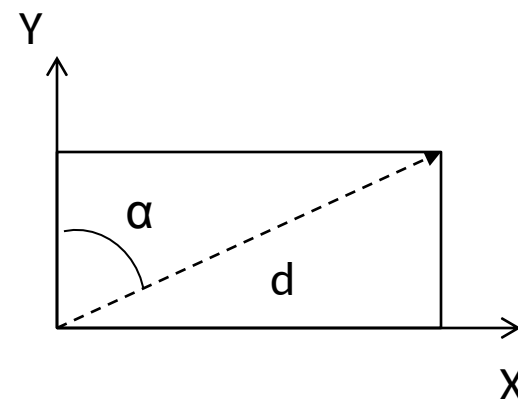
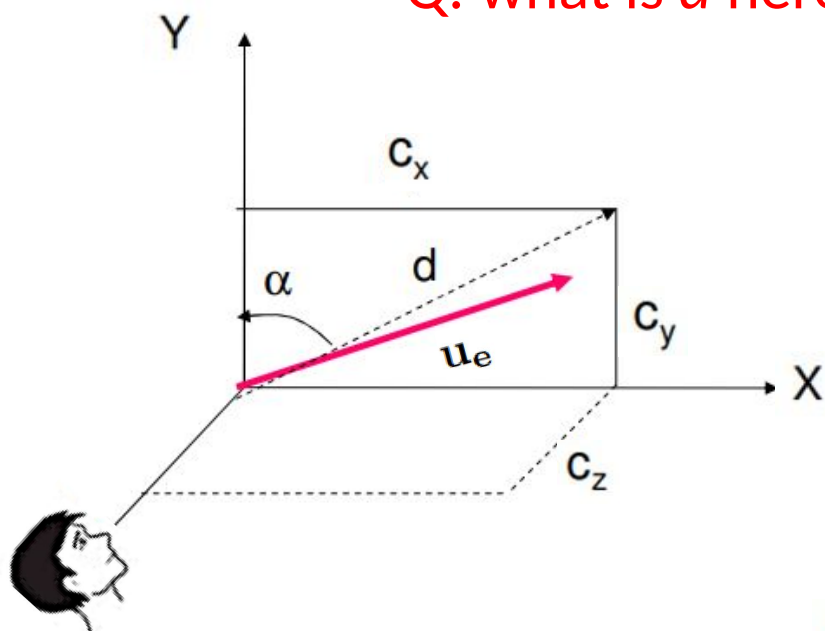
Coinciding the line with Principal axis (2/5)

Coinciding the arbitrary axis with any axis
the rotations are needed about other two axes



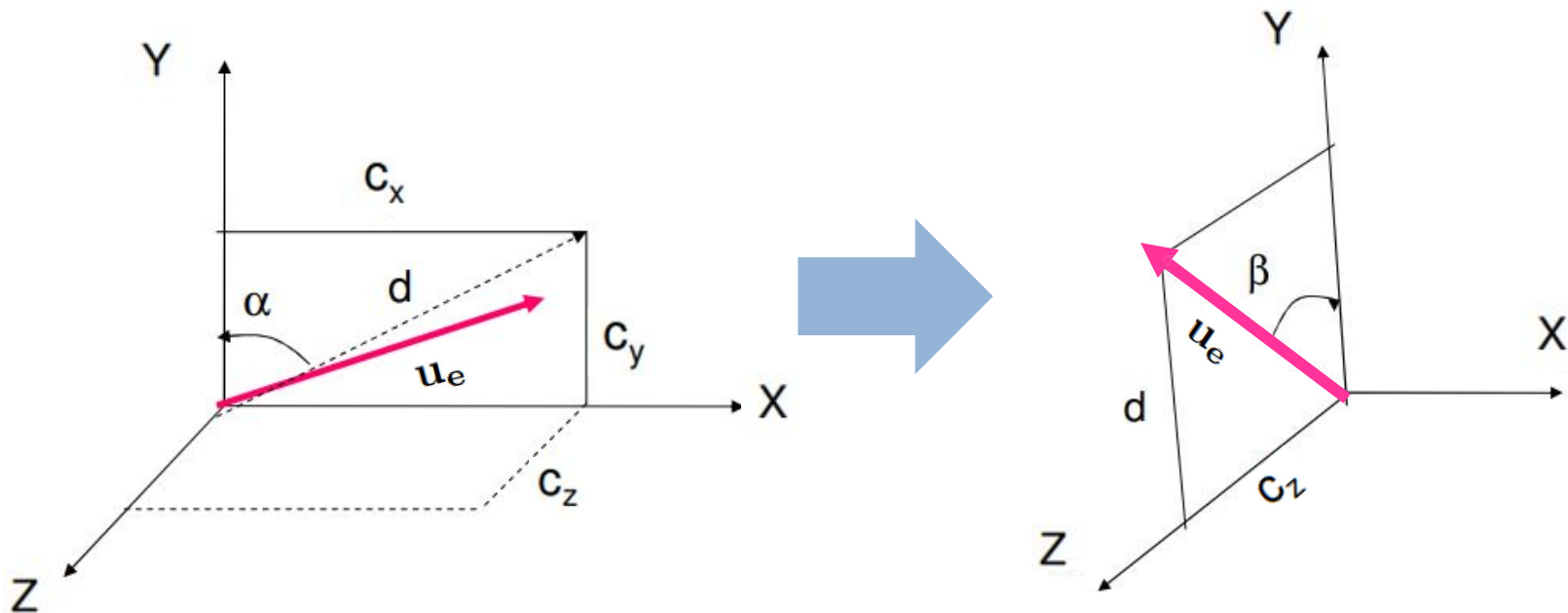
Coinciding the line with Principal axis (3/5)

Q: what is d here?



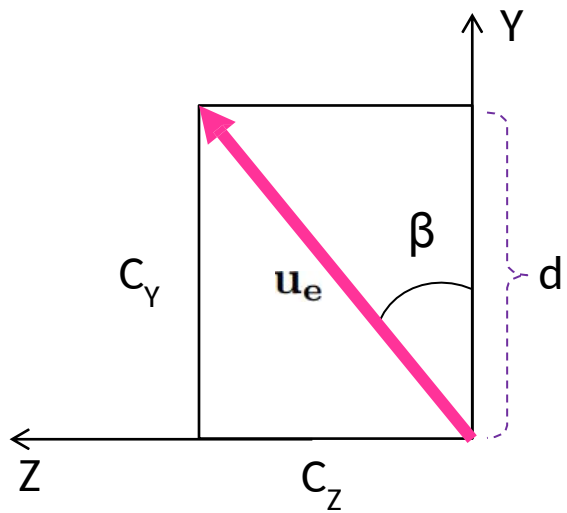
$$d = \sqrt{c_x^2 + c_y^2} \quad \cos \alpha = \frac{c_y}{d} \quad \sin \alpha = \frac{?}{d}$$

Coinciding the line with Principal axis (4/5)

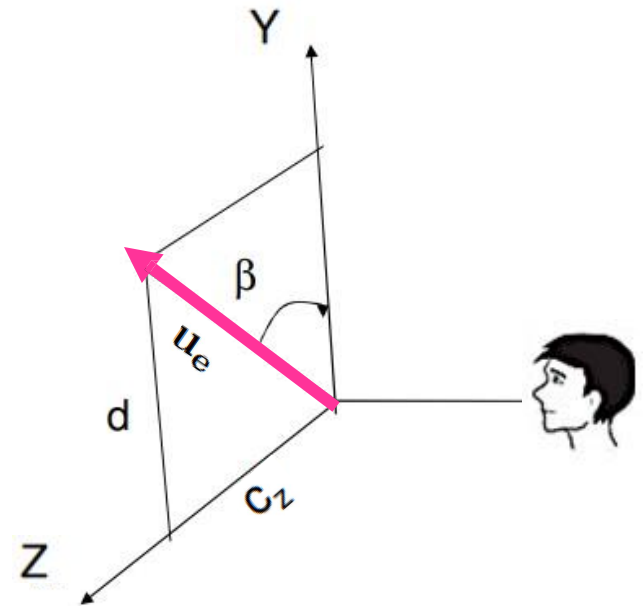


What is the rotation matrix?

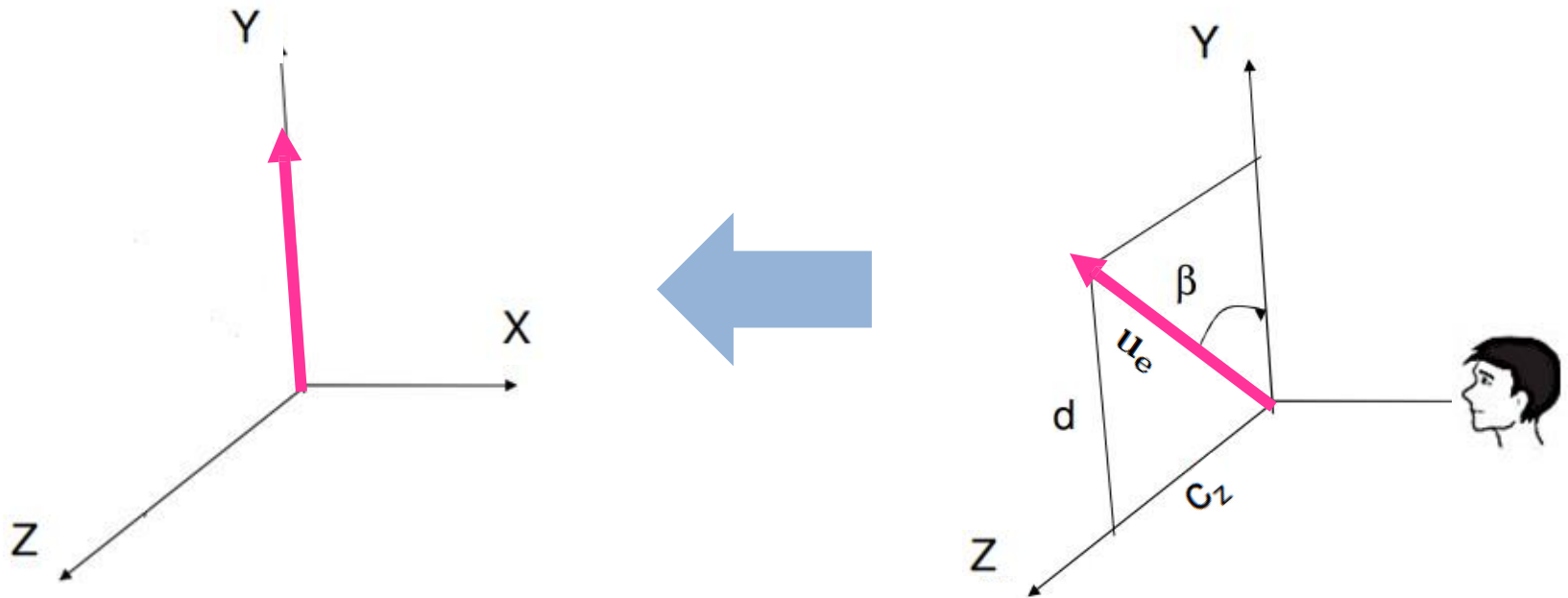
Coinciding the line with Principal axis (5/5)



$$\cos \beta = d \quad \sin \beta = ?$$

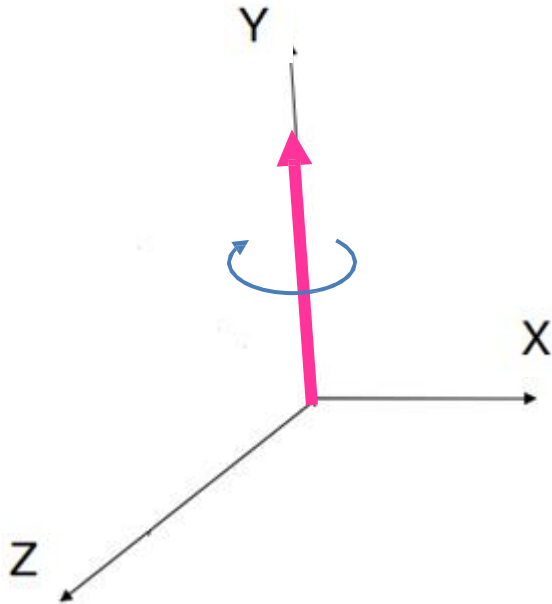


Rotating about the principal axis (1/2)



What is the rotation matrix?

Rotating about the principal axis (2/2)



What is the rotation matrix?

Undoing the steps (1/1)

- Q: What are the undoing steps?

Composite Transformation (1/1)

- $M = T^{-1} * R_x^{-1}(-\beta) * \dots ?$

- $Q' = M * Q$

Practice Problem

- AB is a line and P is a point in 3D space; where the points A,B and P are (1,1,1), (3,3,4) and (2,2,4) respectively. We want to rotate P along AB by +90 degree. Determine the composite transformation matrix to do the task and calculate the rotated point P'.

References

- http://ami.ektf.hu/uploads/papers/finalpdf/AMI_40_from175_to186.pdf
- http://web.iitd.ac.in/~hegde/cad/lecture/L6_3dtrans.pdf